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Anterior cingulate cortex in complex associative learning: monitoring action state and action content

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eLife Assessment

Huang and colleagues examined neural responses in mouse anterior cingulate cortex (ACC) during a discrimination-avoidance task. The authors present **useful** findings that ACC neurons encode primarily post-action variables or "action content" over extended periods. Though the methodological approach was sound, the evidence in support of action state encoding, ruling out alternative explanations related to movement, is **incomplete**.

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Abstract

Environmental changes necessitate adaptive responses, and thus the ability to monitor one's actions and their connection to specific cues and outcomes is crucial for survival. The anterior cingulate cortex (ACC) is implicated in these processes, yet its precise role in action monitoring versus outcome tracking remains unclear. To investigate this, we developed a novel discrimination–avoidance task for mice, designed with clear temporal separation between actions and outcomes. Our findings show that ACC neurons primarily encode post-action variables over extended periods, reflecting the animal's preceding actions rather than the outcomes or values of those actions. Specifically, we identified two distinct subpopulations of ACC neurons: one encoding the action state (whether an action was taken) and the other encoding the action content (which action was taken). Importantly, increased post-action ACC activity was associated with better performance in subsequent trials. These findings suggest that the ACC supports complex associative learning through extended signaling of rich actionrelevant information, thereby bridging cue, action, and outcome associations.

Introduction

The ability to adapt behavior based on environmental cues and outcome-related feedback is essential for survival. Central to this process is the brain's capacity to integrate diverse information to form cue–action–outcome associations that guide future behaviors across varying conditions. However, the neural mechanisms underlying this cognitive flexibility remain poorly understood. The anterior cingulate cortex (ACC) has emerged as a critical node in mediating this process, with growing evidence underscoring its pivotal role in higher-order cognition and adaptive behavior [1–7]. Despite this, the precise role of the ACC in updating and modifying behavior is still under debate [1–7].

Historically, the ACC was thought to be essential for error detection or conflict monitoring [8–10]. However, accumulating evidence has challenged these views [11–17], leading to proposals that the ACC may support a diverse range of functions. These include value encoding, strategy updating, decision-making, action monitoring, and outcome tracking [18–29]. Some of these proposed functions remain controversial, such as the ACC's role in decision-making [18–20], while others are defined more broadly, such as the ACC's role in strategy updating [26, 28]. Nevertheless, there is broad agreement that ACC activity is closely linked to actions and/or action-related outcomes, particularly in tasks involving competing actions [18–29]. Supporting this view, lesions of the ACC impair the maintenance of newly acquired task performance and disrupt behavioral flexibility, including the ability to associate actions with their outcomes [27–30].

Despite evidence linking the ACC to action-related cognitive processes, past studies have primarily relied on simple motor tasks, such as saccades, licking, or joystick movements, leaving its role in more naturalistic behaviors largely unexplored. Moreover, the brief temporal separations between cues, actions, and outcomes in prior studies have made it challenging to determine whether ACC neuronal activity contributes to decision-making, action monitoring, or merely tracking the outcomes of those actions [17–31]. In this study, we aim to disentangle the ACC's role by employing a novel discrimination–avoidance task, designed to evoke robust, naturalistic action responses and establish a clear temporal separation between cues, actions, and outcomes. Our findings highlight a distinct role of the ACC in encoding post-action variables that capture detailed information about preceding actions, rather than tracking the outcomes of those actions or their associated values.

Results

A novel discrimination–avoidance task

We first developed a novel discrimination–avoidance task, tailored to investigate action-based complex associative learning in mice. In this task, animals learn to discriminate between two auditory cues that predict context-dependent footshocks. Specifically, sounds A and B signal electric shocks in rooms A and B of a shuttle box at sound terminations, respectively (Fig. 1A; Fig. S1). This design requires animals to either "stay" in the current room or "shuttle" to the adjacent room during sound presentations to avoid shocks (Supplementary Videos 1–2). During inter-trial intervals, animals are free to explore either room without consequence.

Notably, the auditory cues are not inherently associated with positive or negative valence; instead, their meaning is dynamically determined by the animal's current location (room A or B) at the time of cue presentation, signaling either safety or shock. Thus, this task, which requires discrimination of sensory cues and environmental contexts, as well as the integration of cues, actions, and outcomes, serves as an ideal tool to study complex associative learning. Our results showed that mice gradually learned the task, achieving an average success rate of 76.7% in avoiding shocks by the 10th training session (Fig. 1B). An additional five training sessions yielded a modest improvement, increasing the average success rate to 82.5% on the 15th session (Fig. S2).

Given that the task requires two distinct behavioral responses, we separated the trials into shuttle and stay trials for further analysis. In shuttle trials, mice must move to the adjacent room before the sound ends to avoid shocks. Across training sessions, correct shuttles increased markedly, whereas incorrect shuttles decreased only modestly (Fig. 1C; Fig. S2). This pattern suggests that animals choose to shuttle primarily when they are confident in the outcome (i.e., safety); otherwise, they remain in place. Consistent with this notion, in stay trials, where mice must remain in their current room to avoid shocks, incorrect stays decreased markedly across training sessions, mirroring the improvement in correct shuttle performance. By contrast, correct stays increased only modestly (Fig. 1C; Fig. S2).

We also compared shuttle response latencies between correct and incorrect trials during the late stages of training. On average, the shuttle response latency was 6.5 s during correct shuttles, providing a 3.5-s temporal separation between actions (shuttles) and outcomes (safety or shocks; Fig. 1D). This clear temporal distinction allows us to compare how information is coded during the action *vs.* outcome periods. The longer shuttle latency during incorrect shuttles suggests that last-second responses are more likely to be incorrect (Fig. 1D).

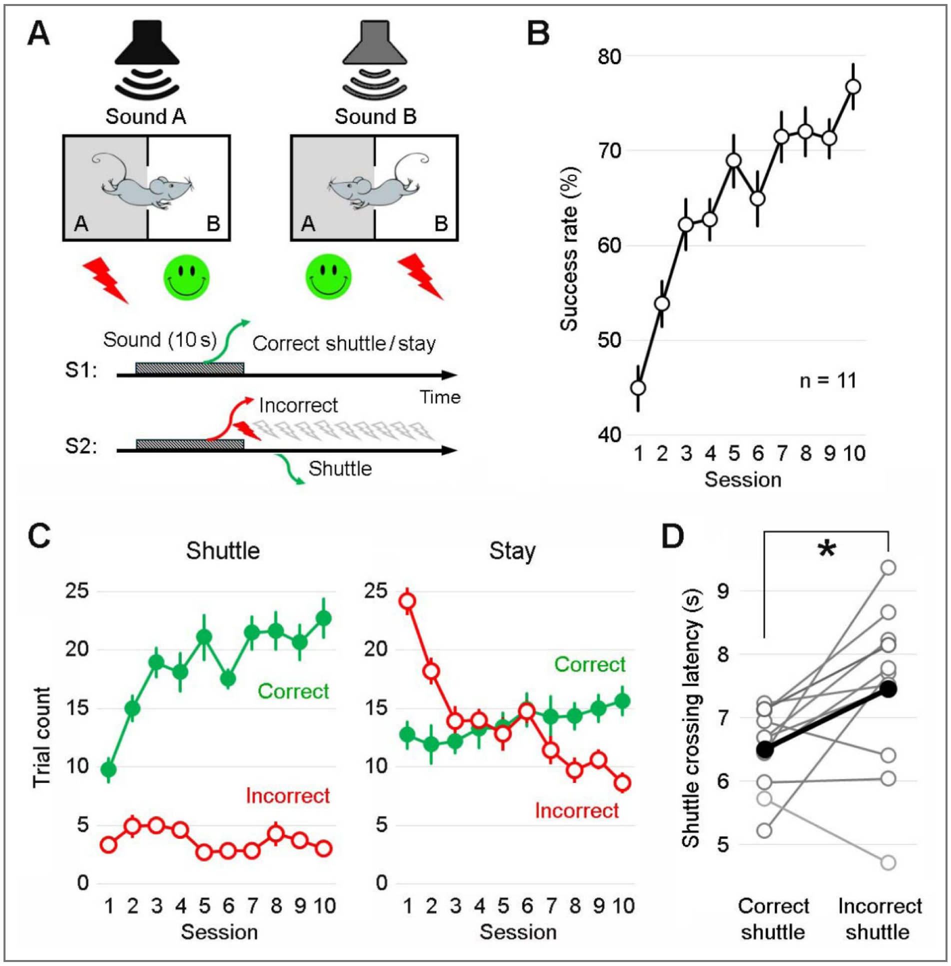


Figure 1. A novel discrimination-avoidance task.

A, Top: Schematic of the task. Mice are trained to discriminate between two auditory cues (lasting 10 s) and shuttle between two distinct rooms of a shuttle box to avoid footshocks. Specifically, sounds A and B signal shocks in rooms A and B, respectively. Bottom: Two behavioral response scenarios. S1: If the mouse makes the correct response, either by staying in or shuttling to the correct room before the sound ends, no shock is administered. S2: If the mouse makes an incorrect response, either by staying in or shuttling to the incorrect room before the sound ends, up to 10 mild shocks (0.5 mA, 0.1 s; 2 s apart) are administered until the mouse shuttles to the correct room. Each training session comprises 50 trials, 60 s apart; sounds A and B are presented in a pseudorandom order. **B**, Learning curve showing the success rate in avoiding shocks across training sessions (mean ± s.e.m.; n = 11 mice). $F_{9, 90} = 14.78$; $P < 0.001$; One-way ANOVA. **C**, Correct and incorrect trial counts for shuttle *vs.* stay trials across training sessions for the same mice shown in B. Correct shuttle: $F_{9, 90} = 14.62$, $P < 0.001$; Incorrect shuttle: $F_{9, 90} = 1.94$, $P = 0.057$; Correct stay: $F_{9, 90} = 1.55$, $P = 0.142$; Incorrect stay: $F_{9, 9\tilde{0}} = 18.73$, $P < 0.001$; One-way ANOVA. **D**, The mean shuttle crossing latency, averaged over the last two sessions for individual mice, is significantly shorter in correct trials than that in incorrect trials ($t_{10} = 2.62$, $P = 0.026$; Effect Size: Cohen's d = 0.789; Power = 0.656; paired *t* test). The black line indicates the mean; grey lines indicate individual mice. Shuttle crossing is defined as the body center crossing the midline opening of the shuttle box.

Next, we conducted multi-channel *in vivo* electrophysiology recordings from the ACC (Fig. 2A; Fig. S2) in mice performing the discrimination–avoidance task after they had completed 10 training sessions and achieved a success rate of 70% or above. We initially observed robust changes in ACC activity following shuttle responses (Fig. 2 B&C). Most ACC neurons exhibited prolonged activity changes lasting up to 30 s, either activation or suppression, primarily after shuttle crossings (Fig. 2 C&E; Fig. S2). In contrast, the same ACC neurons showed no or limited activity changes during stay trials in response to the auditory cues (Fig. 2 D&F).

To assess ACC neuronal population activity dynamics, we performed Principal Component Analysis (PCA) on simultaneously recorded ACC neurons across correct-shuttle and correct-stay trials (Fig. 3A). The first three principal components (PCs) revealed pronounced changes in population activity during shuttle trials, whereas activity remained relatively stable during stay trials (Fig. 3B; Fig. S3). Quantifying the maximum trajectory distance within the 3D PC space revealed consistent, prominent changes in population activity during shuttle responses across animals in five representative sessions (Fig. 3C). The pronounced ACC activity following shuttles, coupled with minimal activity in response to cues, suggests that ACC neurons primarily encode post-action variables. Such sustained post-action activity may support complex associative learning by linking temporally separated action-and outcome-relevant information.

To assess whether ACC activity has a role in encoding pre-action variables, we examined ACC activity aligned to shuttle initiations, defined as locomotion velocity exceeding one standard deviation (s.d.) above baseline (Fig. 4A). Our results revealed diverse response properties of the ACC neuronal population (Fig. 4B). PCA classified these responses into three major categories (Fig. 4 C&D). Pre-shuttle ramping activity was mainly observed in a small subset of neurons, (Types 3 b&c, ~16% of the population), indicating a limited role of the ACC in preaction information coding, including decision-making and planning processes. The remaining ACC neurons primarily exhibited post-shuttle responses, either activation (Types 1a, 1b, 3a) or inhibition (Types 2a, 2b), highlighting the predominant role of the ACC in post-action information processing.

ACC neurons exhibit limited modulation by speed

Given that the post-action ACC activity is often prolonged and outlasts shuttle termination, we hypothesized that this activity is distinct from locomotion encoding. To test this, we analyzed ACC activity in relation to movement speed. Each task session included a 5-min free exploration period before trials begin, allowing us to assess if ACC activity is associated with basic motor functions. We found that only a small portion of neurons showed speed-correlated activity, with either positive (7.7%) or negative correlation (6.9%), indicating that the majority of the neurons are not tuned to movement speed (Fig. 5 A–C). This is consistent with our observation of sustained post-shuttle ACC activity in the absence of movement, which is distinct from locomotion encoding. Nevertheless, it remains unclear whether this small fraction of speed-related neurons represents a distinct subpopulation within the ACC or reflects recordings from nearby motor cortex. Postmortem examination of the recording sites suggests that most neurons were recorded from the ACC, while a small subset located at the border between the ACC and motor cortex (Fig. S2). Therefore, it is possible that the speed-related neurons originated from the motor cortex.

ACC neurons monitor actions independent of outcomes

To determine if the post-action ACC activity encodes information related to outcomes, we analyzed ACC neuronal activity across three distinct conditions: correct shuttles, incorrect shuttles, and post-shock shuttles (Fig. 6A; post-shock shuttles are defined as shuttles following footshocks received during incorrect-shuttle and incorrect-stay trials). Our results revealed that ACC neurons exhibited similar activity patterns across all three conditions, regardless of outcomes (presumed safety *vs.* uncertainty *vs.* safety; Fig. 6B). Specifically, both activity strength (Fig. 6C) and activity pattern (Fig. 6D) were significantly correlated across conditions, indicating outcome-

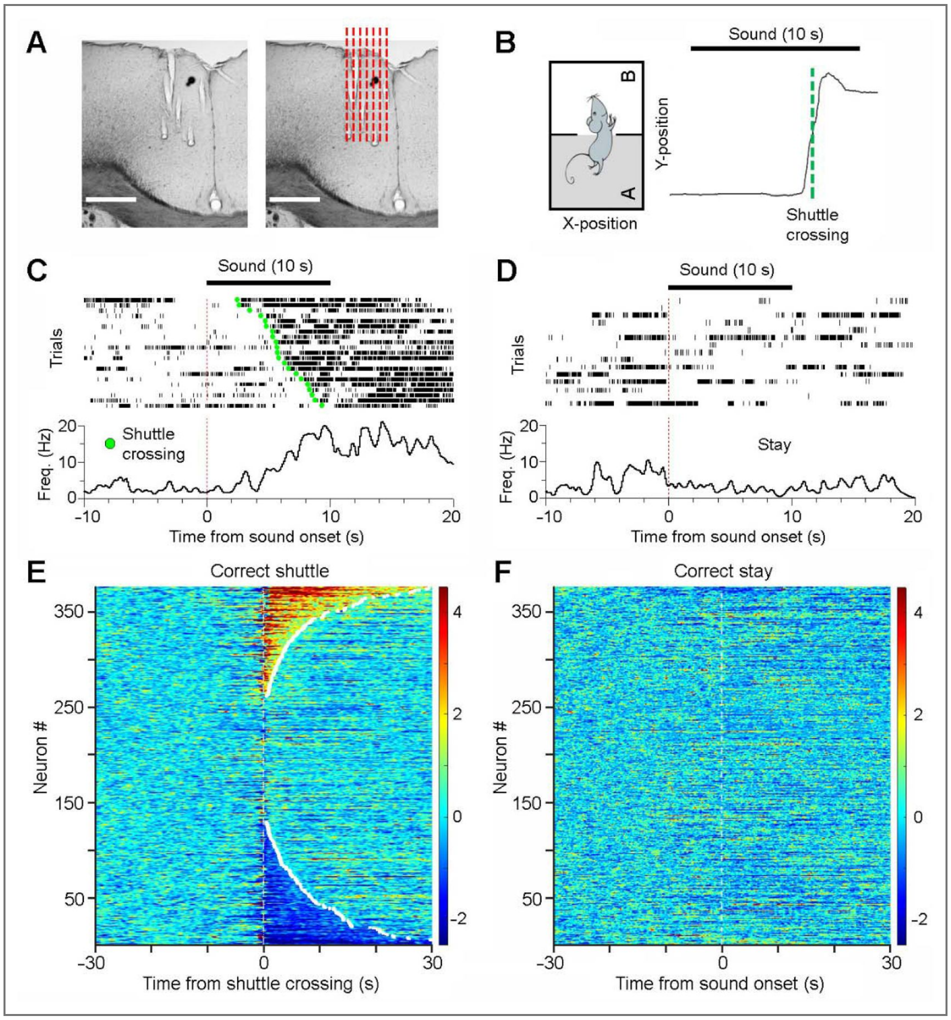


Figure 2. **ACC neurons primarily encode post-action variables.**

A, A representative brain section showing electrode tracks (left) and the presumed implantation sites (red dashed lines; right), located largely within the ACC. Scale bars, 0.5 mm. **B**, A representative shuttle response and corresponding Y-position of the animal's body center. **C&D**, Peri-event rasters and histograms of a representative ACC neuron during correct-shuttle (C; trials sorted by shuttle latency) and correct-stay trials (D). In this session, there were 21 correct shuttles, 15 correct stays, 5 incorrect shuttles, and 9 incorrect stays. **E&F**, Heatmaps showing the activity of all recorded ACC neurons (n = 376) during correct-shuttle (E) and correct-stay trials (F). Neurons in E and F are arranged in the same order.

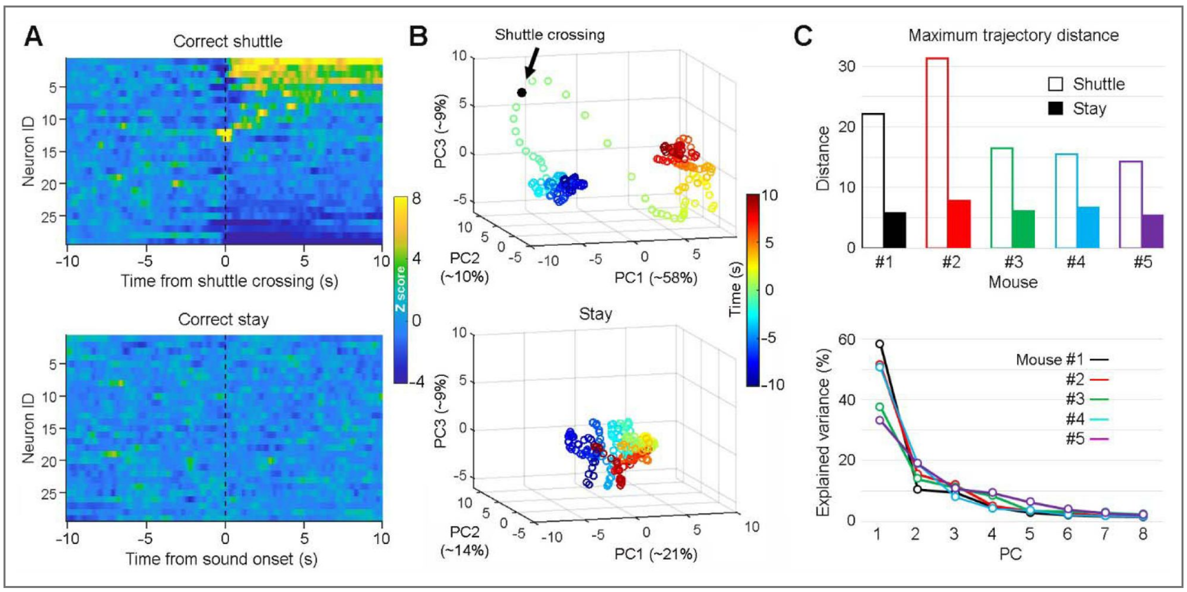


Figure 3. **ACC neurons primarily respond during "shuttle" but not "stay" trials.**

A, Heatmaps showing the activity of simultaneously recorded ACC neurons (n = 29) during correct-shuttle (top) and correct-stay trials (bottom) from the same recording session as shown in Fig. 2C. **B**, Principal component analysis (PCA) of ACC neuronal population activity as shown in A. PC1, PC2, and PC3 are the first three principal components; the numbers are the percentages of total variance explained by the corresponding PCs. Each circle indicates a time lapse of 0.1 s. Note that there is a robust neural state change in the 3-D PC space surrounding the shuttle response (top), but not the stay response (bottom). **C**, Maximum trajectory distance between the center of the baseline period and any post–Time 0 data point, as shown in B, across five animals (top), and the corresponding scree plots showing variance explained by the first eight PCs during shuttle trials (bottom).

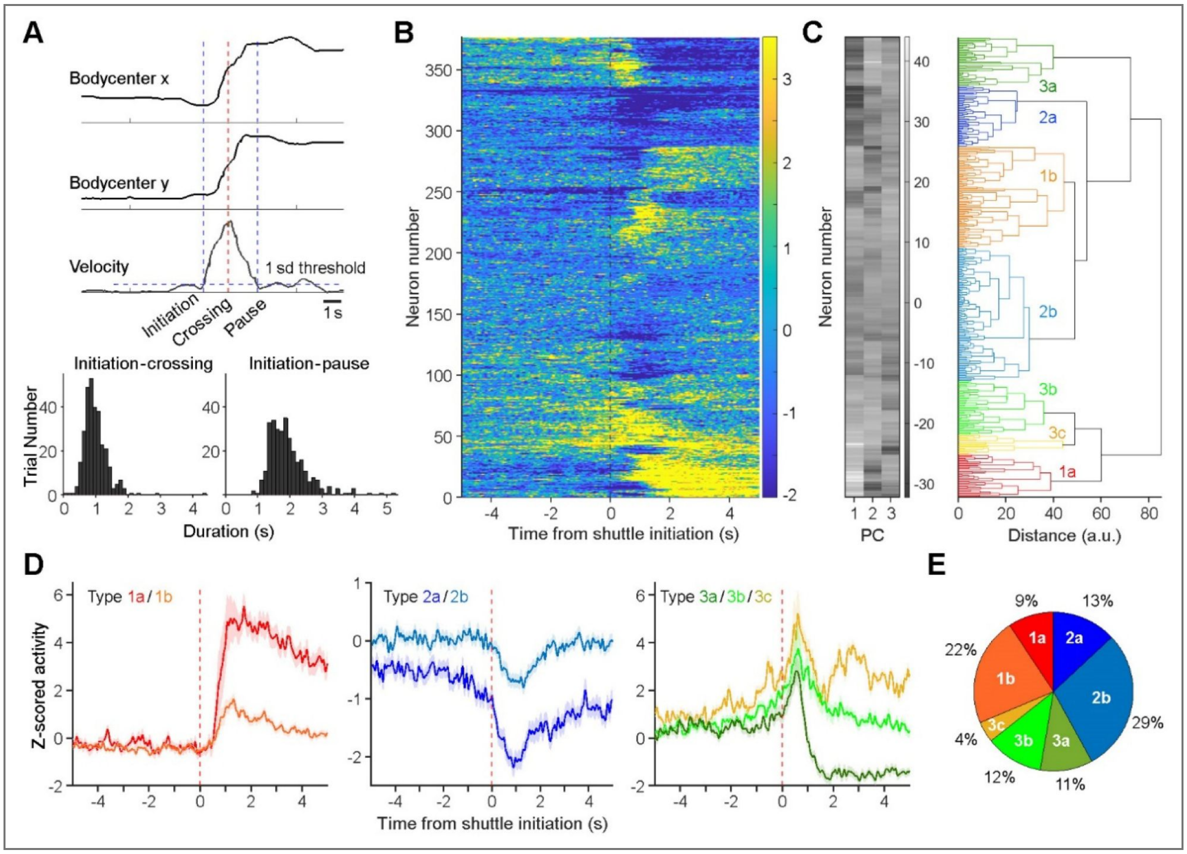


Fig. 4. Characterizing ACC neuronal activity in relation to action initiations.

A, Top, schematic illustrating the definition of shuttle initiation, crossing, and termination (pause). Bottom, distributions of half-(left) and full-shuttle durations (right). **B**, Z-scored activity of all ACC neurons (n = 376) during correct shuttle trials. **C**, Principal-component analysis (PCA) classifies ACC neuronal activity (as shown in B) into seven categories. PC1, PC2, and PC3 represent the first three principal components color coded from low (dark) to high scores (white). **D**, Mean activity (± s.e.m.) of the seven categories of ACC neurons. **E**, Fractions of individual categories of ACC neurons.

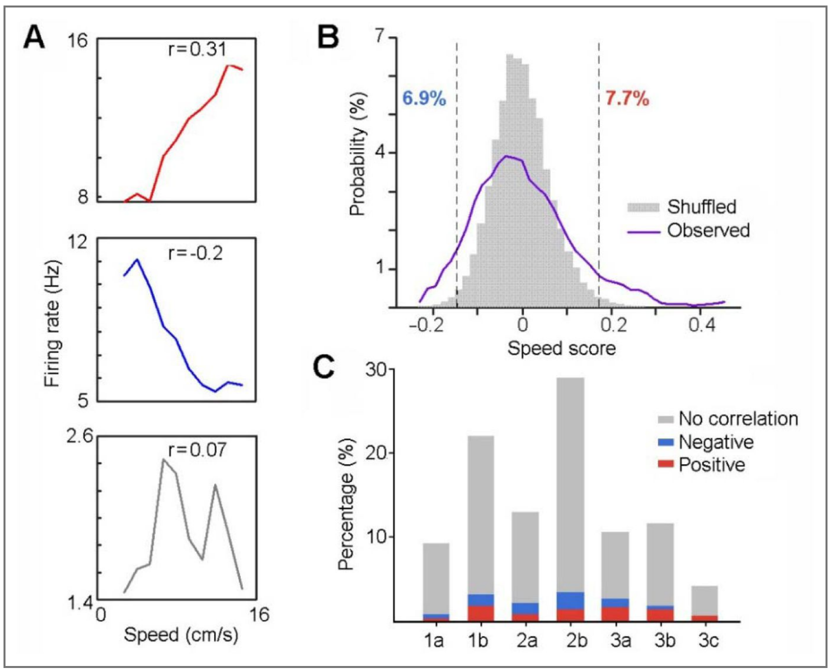


Figure 5. **ACC neurons exhibit limited modulation by speed.**

A, Speed-tuning curves of three simultaneously recorded neurons during free exploration in shuttle boxes, showing positive (red), negative (blue), or no correlation (gray) between firing rate and locomotion speed. "r" indicates Pearson's correlation coefficient. **B**, Distributions of observed *vs.* shuffled correlation values across all recorded ACC neurons (n = 376). Dashed lines mark the 99th percentile of shuffled distribution. Overall, 7.7% of recorded neurons were positively modulated by speed, and 6.9% were negatively modulated. **C**, Proportions of speed-modulation for each category of ACC neurons (see Fig. 4D).

independent responses in ACC neurons. Consistently, we found that ACC neurons showed limited responses to footshocks during incorrect trials (Fig. S4). Together, these findings suggest that ACC neurons monitor actions independent of outcomes or values associated with these actions.

ACC neurons monitor *action state* and *action content*

Since post-action ACC activity was outcome-independent, we next asked which other features of the task ACC activity might differentiate. Leveraging the two distinct shuttle responses within the task, we examined whether ACC neurons responded differently between rooms A→B shuttles (in response to sound A) and rooms B→A shuttles (in response to sound B). Our analyses identified two major groups of ACC neurons based on their responses to these shuttles. The first group showed indiscriminate response, exhibiting either increased or decreased activity without differentiating between A→B and B→A shuttles (Fig. 7: Neurons 1&2). We propose that these ACC neurons encode an *action state*, a variable representing the change of behavioral state in response to the auditory cues.

In contrast, the second group of ACC neurons exhibited discriminate responses between rooms A→B and B→A shuttles. These neurons either responded selectively in one set of shuttles or show different levels of responses (activation or inhibition) between the two sets of shuttles (Fig. 7: Neurons 3&4; Fig. S5). We propose that these ACC neurons encode *action content*, a variable representing distinct action information (i.e., shuttling from rooms A→B *vs.* B→A).

To quantify *action-state* and *action-content* neurons, we performed a generalized linear model (GLM)-based analysis of ACC activity surrounding shuttle responses (see Methods). Based on coefficient β or Δβ value differences, most ACC neurons were classified as either *action-content* neurons (54.5%; Categories 1&2) or *action-state* neurons (19.1%; Categories 3&4; Fig. 7 C-E). These findings suggest a prominent role for post-action ACC activity in encoding rich information about preceding *action state* and *action content*.

Notably, ACC activity does not resemble place cell activity observed in the hippocampus [32], as evidenced by our analysis of spike activity from the intertrial-interval periods (Fig. S6). This aligns with prior findings that ACC neurons do not simply encode spatial information [33], although *action-content* ACC neurons likely incorporate spatial variables, given their shuttleroute response selectivity.

To determine if the post-action ACC neuronal population activity can decode shuttle contents (rooms A→B *vs.* B→A shuttles), we implemented a machine-learning approach. Specifically, we trained binary support vector machine (SVM) classifiers and performed cross-validations (Fig. 8A). Our results revealed that the post-shuttle population ACC activity was highly effective in decoding the animal's preceding actions of either A→B *vs.* B→A shuttles, reaching a decoding accuracy close to 90% on average (Fig. 8B). Although the pre-shuttle ACC activity can also decode the shuttle content, it did so with much lower accuracy (Fig. 8B). These findings provide compelling evidence supporting post-action ACC activity in encoding detailed information about the animal's preceding actions. Further analysis confirmed that model prediction accuracy was derived primarily from *action-content* ACC neurons, as gradual removal of these neurons from the analysis resulted in progressive decreases in prediction accuracy (Fig. 8C).

Post-action ACC activity influences future performance

We next investigated whether post-action ACC activity following a trial influenced performance on the subsequent trial, which occurred ~1 min later. Specifically, we divided correct-shuttle trials into two groups (Fig. 9A): those followed by correct trials (including both correct stays and shuttles), and those followed by incorrect trials (including incorrect stays and shuttles). Our results revealed that post-shuttle ACC activity was notably higher when it preceded correct trials rather than incorrect ones (Fig. 9B). *The strength of post-shuttle ACC activity may reflect task engagement*, with greater engagement facilitating learning. Statistically, the top one third of the most responsive ACC neurons exhibit significantly higher activity that preceded correct trials than

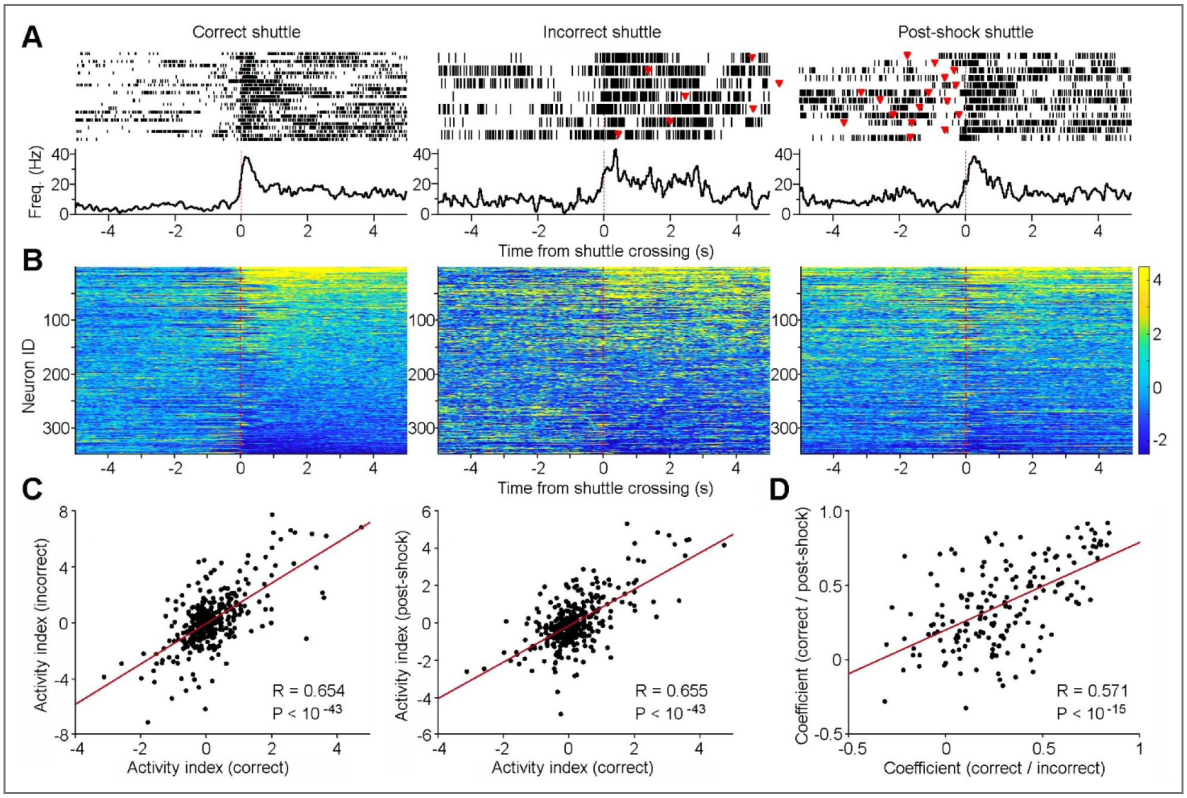


Figure 6. ACC neurons monitor actions independent of outcomes.

A, Peri-event rasters (trials) and histograms of a representative ACC neuron during correct (left), incorrect (middle), and post-shock shuttles (right) within a session. Red triangles indicate shock administrations. Note that incorrect shuttles are followed by a second shuttle after animals receive footshocks, and approximately half of the postshock shuttles are preceded by incorrect shuttles. **B**, Heatmaps showing the activity of individual ACC neurons (n = 348) during correct (left), incorrect (middle), and post-shock shuttles (right). Neurons are arranged in the same order across the three heatmaps. Color bar indicates z-scored activity. Note that the number of incorrect shuttles in a session is often <7, leading to greater variability in mean activity. Only sessions with >4 incorrect shuttles are included in the analysis. **C**, Activity indexes of individual ACC neurons between correct and incorrect shuttles (left), and between correct and post-shock shuttles (right). Activity index is defined as: Activity Index = $\text{Mean}^{\text{post-huttle}}$ – $\text{Mean}^{\text{pre-huttle}}$, where $\text{Mean}^{\text{pre-huttle}}$ and $\text{Mean}^{\text{post-huttle}}$ are the mean z scores calculated between -5-0 and 0-5 s, respectively, as shown in B. **D**, Correlation coefficients of the activity (−5-5 s) between correct and incorrect shuttles (x axis) and between correct and post-shock shuttles (y axis). Only the top and bottom quartiles of ACC neurons (as shown in B) are used for the analysis.

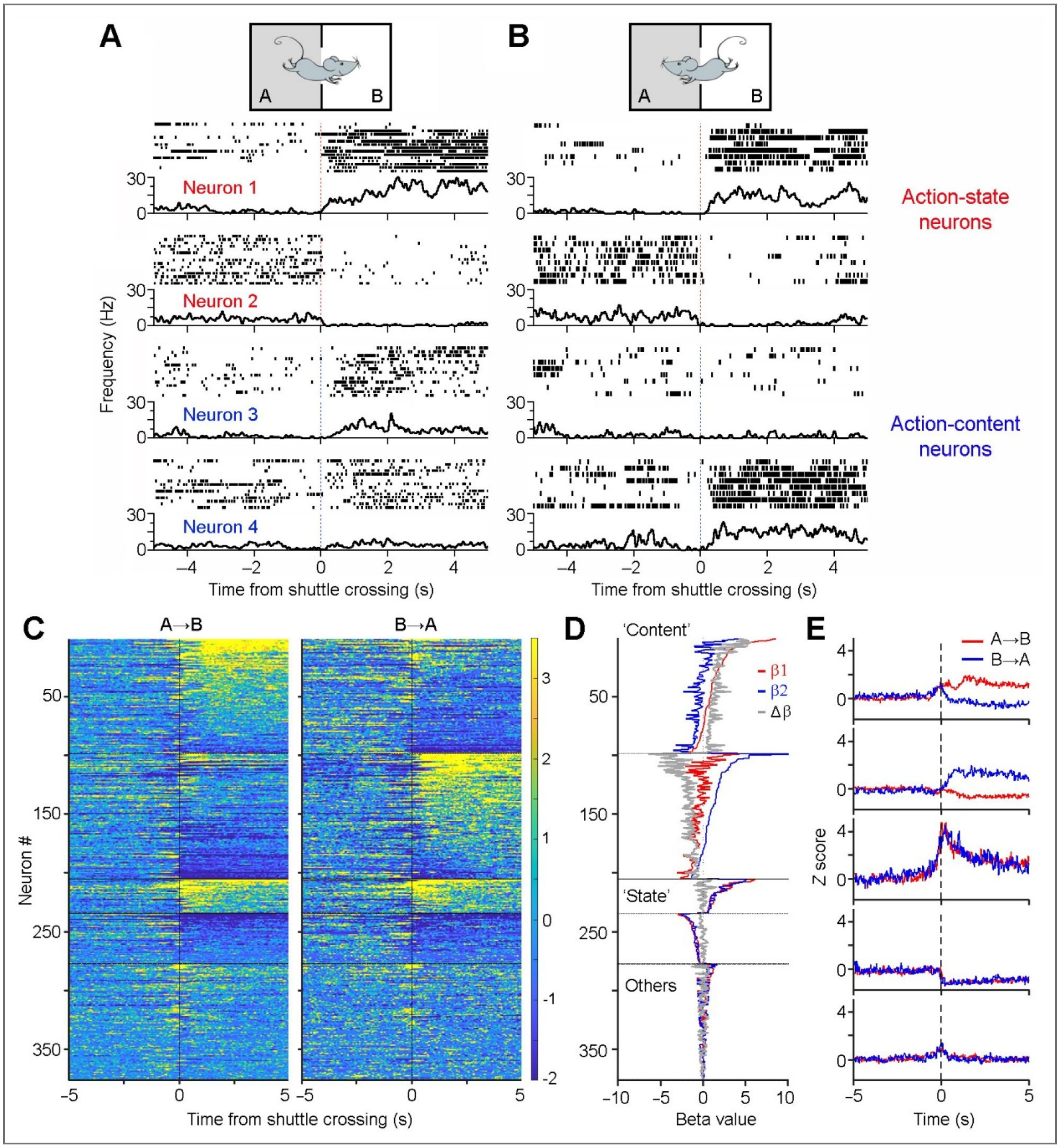


Figure 7. **ACC neurons monitor *action state* and *action content*.**

A&B, Peri-event rasters (trials) & histograms of four simultaneously recorded ACC neurons during two sets of shuttles: rooms A B shuttles (A) *vs.* rooms B A shuttles (B). Notably, neurons 1&2 exhibit indiscriminate responses, either increasing or decreasing their activity after shuttles, thereby monitoring *action state* changes. In contrast, neurons 3&4 are selectively activated in one set of the shuttles, thereby monitoring *action content* (i.e., rooms A B *vs.* B A shuttles). **C**, Z-scored activity of all ACC neurons (n = 376) during A B shuttles (left) and B A shuttles (right). The color bar indicates z score. Neurons are categorized and sorted according to coefficient β or Δβ values (see Methods): Category 1 (n = 98), $\beta_1 > \beta_2$, sorted by β_1; Category 2 (n = 107), $\beta_1 < \beta_2$, sorted by β_2; Categories 3&4 (n = 29/43), both β_1 and β_2 are significantly positive or negative, sorted by the combined magnitude of β_1 and β_2. **D**, Coefficient values β_1 β_2, and Δβ (β_1–β_2) values are shown in red, blue, and gray, respectively, for individual ACC neurons ordered in the same sequence as shown in C. **E**, Mean activity for the five major neuronal categories (as discussed in C) during A B (red) and B A (blue) shuttles.

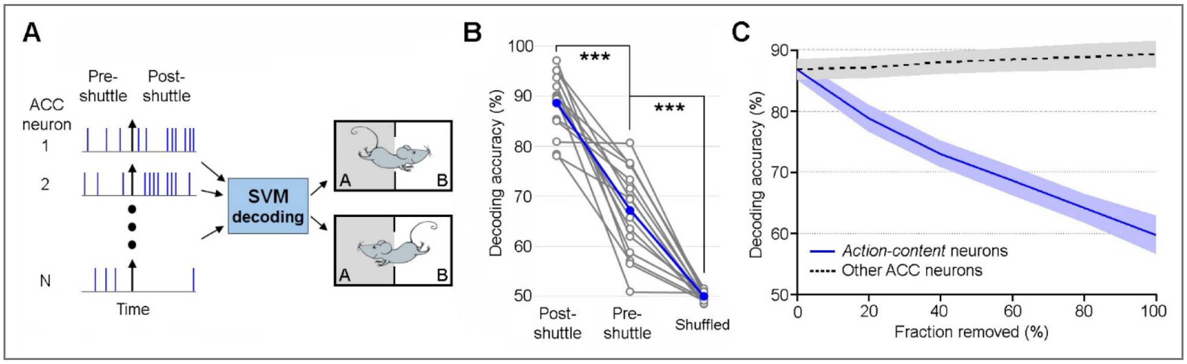


Figure 8. **Post-shuttle ACC neuronal population activity decodes *action content*.**

C, Schematic diagram of support vector machine (SVM) decoding. ACC neuronal population activity from pre-shuttle period (−5–0), post-shuttle period (0–5), or shuffled spikes is used to train the decoder and subsequently distinguish between action content (rooms A B *vs.* B A shuttles). **B**, Mean decoding accuracy (blue line) and individual decoding accuracies for 15 sessions (grey lines). *Friedman test (*P < 0.001*)* and *post-hoc Wilcoxon signed-rank test with Bonferroni correction* (***P < 0.001). **C**, SVM decoding accuracy across all 15 sessions as a function of the fraction of neurons removed (20% per step), applied separately to *action-content* neurons (blue) or the remaining neurons (grey) within each session (P < 0.001 for each comparison between the two removals; *Wilcoxon signed-rank test*). Shaded areas denote ± s.e.m.

incorrect ones (Fig 9C). This correlation between higher ACC activity and future correct performance suggests that post-action ACC activity may contribute to trial-to-trial behavioral adjustment that underlie associative learning [34].

As a control, we also examined whether the status of the preceding trial influenced post-action ACC activity in the current trial. Specifically, we divided correct-shuttle trials into two groups: those preceded by correct trials, and those preceded by incorrect ones (Fig. 9D). Our results showed no difference in post-shuttle ACC activity between the two conditions (Fig. 9 E&F). This finding was expected, as both positive and negative reinforcement can similarly contribute to associative learning and neural plasticity.

Discussion

Our newly designed discrimination–avoidance task is unique in that it allows us to disentangle the roles of ACC neurons across several proposed functions, including value encoding, decision-making, action monitoring, and outcome tracking. First, this task requires animals to discriminate both sensory cues and environment contexts. Unlike established tasks that often assign fixed positive or negative values to cues, the cues in our task are not inherently associated with valence. Instead, their meaning is dynamically determined by the animal's location (context) at the time of cue presentation. By removing valence from the cues, this design helps disentangle the ACC's potential role in value encoding from other cognitive functions. Second, this task involves robust, ethologically relevant actions (i.e., shuttles), unlike many established paradigms that rely on less naturalistic behaviors such as saccades or lever presses. Finally, the clear temporal separation between actions and outcomes helps disentangle the ACC's roles in action monitoring *vs.* outcome tracking.

Utilizing this task, we find that the ACC primarily encodes post-action variables. Specifically, ACC neurons exhibit robust post-shuttle responses across various conditions, including correct, incorrect, and post-shock shuttles. Despite conditions signaling distinct outcomes (presumed safety *vs.* uncertainty *vs.* safety), the response properties of the ACC neurons remain consistent. Moreover, very few ACC neurons respond directly to positive outcomes (safety) or negative outcomes (shocks). Together, these findings indicate that post-shuttle ACC activity primarily monitors animal's most recent actions, rather than tracking the outcomes or values associated with those actions [21,29, 30].

Our results further reveal two distinct groups of ACC neurons that encode different aspects of actions: *action state* and *action content*. Action state appears to represent the animal's preceding choice of whether an action was taken, updating changes in behavioral state within the ongoing task. In contrast, *action content* represents the specific actions taken (i.e., shuttles from rooms A→B *vs.* B→A), preserving detailed action information. The response selectivity of *action-content* ACC neurons reinforces the notion that ACC activity monitors preceding actions rather than action-associated outcomes or values, which are uniform across these actions. Notably, our findings do not support an alternative interpretation that *action content* reflects correct *vs.* incorrect shuttles. Theoretically, categorizing actions solely as correct or incorrect may not be necessary, as both positive and negative outcomes can guide future actions and contribute to learning [30]. Potential neural networks mediating the sustained post-action ACC activity include thalamic and retrosplenial inputs, given the well-established roles of thalamocortical circuits in sustaining cortical activity [35, 36] and the retrosplenial cortex in processing spatial and directional information [37–40].

Our study also reveals that ACC neurons play a limited role in encoding pre-action variables associated with decision-making or planning, as evidenced by their minimal responses to auditory cues and the modest activity changes prior to shuttle initiation. These findings align with recent research, showing that ACC neurons are mainly involved in post-decisional information processing rather than decision-making itself [18–20]. Nevertheless, substantial evidence from other studies supports the ACC's involvement in value coding or value updating, as reflected in its differential responses to discriminative sensory cues that precede decisions [25, 31, 41]. One possible explanation for the discrepancy in our findings is that the sensory cues used in our task

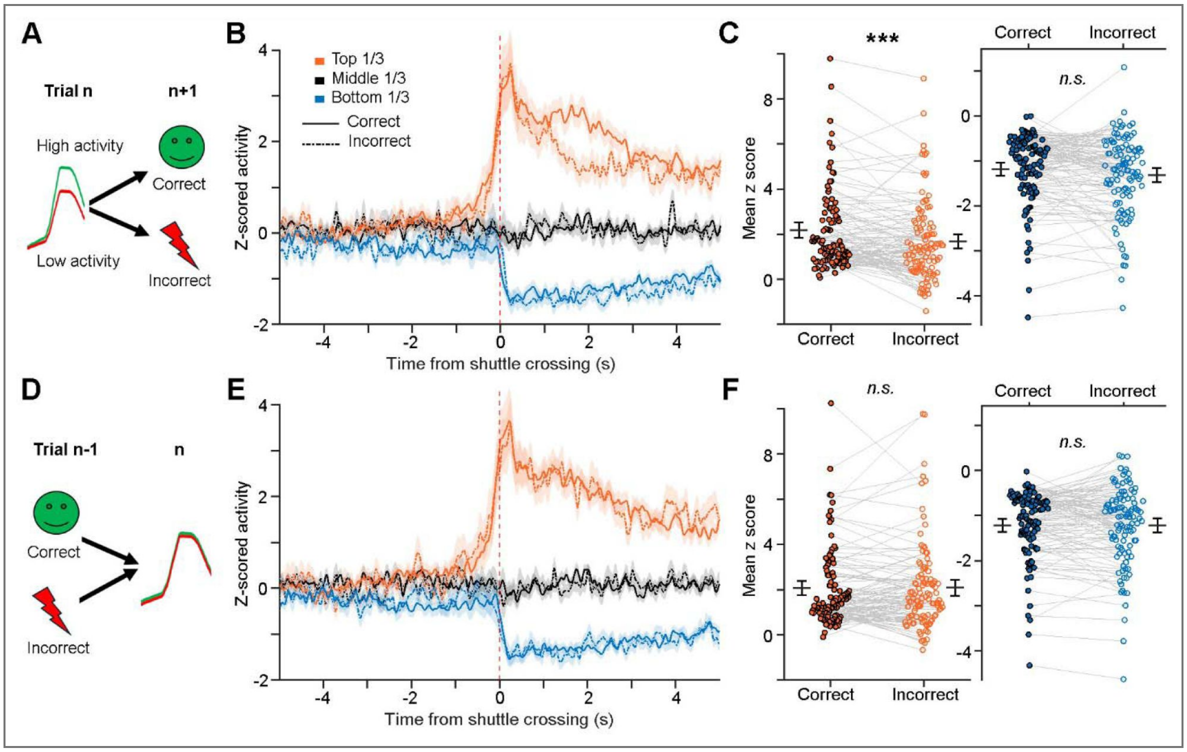


Figure 9. **Post-action ACC activity influences future performance within a task session.**

A, Schematic illustration. **B**, Mean activity (± s.e.m.) of post-shuttle activated neurons (orange lines; top 1/3), inhibited neurons (blue lines; bottom 1/3), and remaining ACC neurons (black lines; middle 1/3), which preceded either correct trials (solid lines) or incorrect trials (dashed lines). **C**, Further comparison of the activation strength for post-shuttle activated neurons (left) and inhibited neurons (right) between the two conditions. Each pair of dots indicates an ACC neuron. Two-way ANOVA, Interaction: $F_{2, 328} = 5.69$; $P = 0.004$; Simple effect for Top1/3: ***$P < 0.001$; Effect Size: Cohen's d = 0.33. **D–F**, Similar to A–C, except that the comparison is based on the status of the preceding trials. Mean z scores in C and F were calculated between 0–5 s after shuttle crossings. Two-way ANOVA revealed no-significant difference. n.s., non-significant.

are not intrinsically linked to specific values. Although these cues play a crucial role in driving go (shuttle)/no-go (stay) decisions, their values are dynamic and vary across trials. This lack of value association may explain why we observe minimal response of ACC neurons to these cues. In contrast, the ACC activity reported in previous studies likely reflects the stable value of the cues [31, 42, 43]. It remains to be determined if distinct ACC subpopulations may be responsible for value encoding *vs.* action monitoring, or if the same ACC neurons multiplex across these functions.

Our results suggest that ACC activity is not directly associated with locomotion. First, both *action-state* and *action-content* neurons tend to show sustained activity even when the animals remain immobile after completing shuttle behaviors, suggesting that their activity is not driven by locomotion. Furthermore, *action-content* neurons are selectively engaged in only one of the two shuttle categories, either rooms A→B or B→A shuttles. Therefore, differences in neuronal activity are unlikely to reflect locomotor differences, given that both shuttle types involve similar movement patterns. Finally, we show that only a small fraction of neurons (14.6%) exhibits locomotion speed-correlated activity. Overall, these results suggest that post-action ACC activity reflects information about preceding actions, independent of the animal's current movement.

One caveat of our study is that the discrimination–avoidance task requires weeks of training in mice. By the time they master the task, ACC activity may reflect modified neural circuits. Investigating ACC activity during early phase of learning, such as by introducing a new pair of cues or contexts, could provide further insights into ACC's role in learning and cognitive processes. Additionally, previous studies have highlighted ACC's key role in reversal learning [44–47]. Future research examining how ACC neurons respond when task rules are reversed could provide further insight into this function. Finally, a limitation of the current study is the lack of evidence for the causal role of post-action ACC activity in complex associative learning. Future investigations using closed-loop strategies to selectively disrupt ACC activity during the post-action phase could help address this question.

Lastly, our findings suggest that post-action ACC activity plays a key role in shaping future behavior. Specifically, we find that increased post-action ACC activity is linked to future performance in subsequent trials, highlighting the ACC's role in facilitating learning and guiding behavior. The level of post-shuttle ACC activity may reflect task engagement, with greater engagement facilitating learning and improving future performance. This aligns with previous research demonstrating the ACC's critical role in complex and flexible learning, including discriminative avoidance learning [48], discriminative extinction learning [49], reversal learning [44–47], task switching [25], trial-to-trial behavioral adaptation [34], and action–outcome associative learning [28–30]. We speculate that both *action-state* and *action-content* ACC neurons contribute to complex associative learning through extended signaling of action-relevant information, thereby bridging cue, action, and outcome associations (Fig. 10). Specifically, *action-state* ACC neurons signal whether an action was taken, while *action-content* ACC neurons provide specific details about what the action was—the content. This action-related information, eventually, is integrated with cue-and outcome-related variables to form complex cue–action–outcome associations that guide future behavior (Fig. 10). Without the ACC, action-relevant information could be lost after action execution, thereby disrupting cue– action–outcome associative learning. Given our key finding that the ACC primarily encodes action-related variables rather than cue- or outcome-related information, we speculate that the integration of these complex associations takes place in other higher cognitive areas, such as the prelimbic cortex and other medial prefrontal subregions [50–52].

Methods

Mice

Male C57BL/6 mice (Jackson Laboratory, stock #000664) were used in this study. The mice were 8–10 weeks old at the start of discrimination–avoidance task training. All mice were group-housed (2–4 mice per cage; 40 × 20 × 25 cm) with corn cob bedding and cotton nesting material, except that after electrode implantation surgery, mice were singly housed. They were maintained on a 12-h

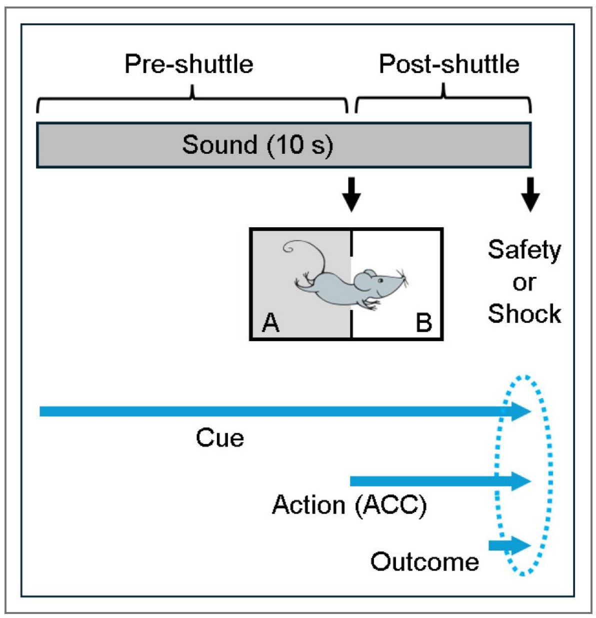


Figure 10. **Proposed model of cue-action-outcome associative learning.**

Top: Four distinct phases of information coding: pre-shuttle, shuttle, post-shuttle, and outcome phases. **Bottom:** Integration of cue, action, and outcome information occurs when all three components are concurrently available (indicated by the oval). Notably, without the ACC, action-relevant information could be lost after action execution, thereby disrupting cue–action–outcome associative learning.

light/dark cycle with *ad libitum* access to food and water. All experimental procedures were approved by the Institutional Animal Care and Use Committees at Drexel University and adhered to the National Research Council's Guide for the Care and Use of Laboratory Animals.

Sounds and shuttle box

Two 10-s auditory cues, 5-kHz tone at ~75 dB and white noise at ~65 dB, were chosen for sound discrimination. Both sounds included 50-ms shaped rise and fall times to reduce abruptness and minimize potential startle responses. The shuttle box used in the experiment was a square chamber measuring 25 × 25 × 32 cm, with a 36-bar shock grid floor, illuminated by lights inside sound-attenuating cubicles (64 × 75 × 36 cm) equipped with speakers (*Med Associates*). The shuttle box was divided at the midline by a plastic divider into two rooms. These two rooms were slightly modified for discrimination purposes: one had two black walls, one white wall, and one transparent wall, while the other room had two white walls, one metal wall, and one transparent wall (Fig. S1). The divider had a 2-inch opening in the center to allow the mice to move freely between the rooms. Animals' behaviors were recorded using Video Freeze software (*Med Associates*) [53].

Discrimination–avoidance task

Prior to training, all mice underwent two daily handling sessions (~10 min each). Once training began, the mice received one training session per day, 5 days per week. In the task, the mice were trained to discriminate between two distinct sounds (A and B) and shuttle between two adjacent rooms (A and B) within a shuttle box to avoid potential footshocks. Specifically, sounds A and B signaled electric shocks in rooms A and B, respectively (Fig. 1A). During training, the mice were first allowed to freely explore the shuttle box for 2 min before trials began, except on the first day of training, where the free exploration period was extended to 10 min.

The training procedure consisted of two phases: pre-training and training. Pre-training phase: mice underwent five daily sessions of sound alternation trials 5/5 (AAAAABBBBB …). Mice that did not exhibit shuttling responses during the first session were excluded from further training. Training phase: mice underwent 10 daily sessions (5 sessions per week) of alternation trials in a pseudorandom order (ABBABAAB .).

Each training session comprised 50 trials, 60 s apart. On incorrect trials, up to 10 scrambled electric shocks (0.5 mA; 0.1 s) were administered starting at sound terminations and continued for an additional 18 s (1 shock every 2 s), except during the pre-training phase, where up to 20 scrambled electric shocks were administered. Shocks were terminated once the mice navigated to the adjacent safe room. The mouse's success rate in avoiding shocks at sound terminations was defined as: Success rate (%) = (Correct stays + Correct shuttles) / All trials.

Real-time location detection

We employed MATLAB functions to detect animal location and subsequently control shock administration (Fig. S1). Specifically, *Med Associates* Video Freeze software recorded real-time video footage of the animal's behaviors [53]. During each trial, MATLAB captured screenshots of the ongoing video at sound terminations and every 2 s thereafter during the shock period. MATLAB then performed background extraction on the captured images to determine which room the mouse was located in. To deliver a shock, MATLAB sent a "shock" signal to an *Arduino UNO* circuit board, which relayed that signal to *Med Associates*, triggering shock administration.

Stereotaxic surgery

Mice that had completed 10 training sessions and surpassed a success rate of 70% during discrimination-avoidance tasks were used for surgery. In brief, mice were anesthetized with ketamine/xylazine mixture (~100/10 mg/kg, i.p.) and maintained on a heating pad at 37°C. Following that, mice received an intra-ACC implantation of a custom-made electrode array (8 tetrodes) [54, 55], and the implant was secured to the skull with stainless screws and resin ionomer (*DenMat*). The coordinates used were AP 1.0 mm, ML 0.4 mm, and DV 1.0 mm.

In vivo recording during discrimination–avoidance tasks

We used tetrodes for recording [54, 55]. Each tetrode consisted of four wires (90% platinum 10% iridium; ~18 μm diameter; *California Fine Wire*). Neural signals were preamplified, digitized, and recorded using a *Blackrock Neurotech* CerePlex, while the animals' behaviors were simultaneously recorded. Spikes were digitized at 30 kHz and filtered between 600–6000 Hz. The recorded spikes were manually sorted using *Plexon* Offline Sorter [56, 57], with key datasets verified by a second experimenter. Tetrode arrays were gradually lowered through a microdrive [58, 59] to record at multiple depths within the ACC, with 2–4 depths from each animal (DV ~1.0–1.3) used for analysis.

Overall, ACC spikes from five mice across 15 sessions were analyzed, with neuron counts of 20/23/29/32 (mouse #1; 4 sessions), 17/20/22 (mouse #2; 3 sessions), 28/24 (mouse #3; 2 sessions), 33/25/21/27 (mouse #4; 4 sessions), and 22/25 (mouse #5; 2 sessions), respectively. The total number of correct/incorrect shuttles used for analysis are 19/5, 19/4, 21/5, 20/4 (mouse #1); 20/7, 23/7, 20/7 (mouse #2); 19/4, 16/2 (mouse #3); 26/4, 23/4, 17/6, 25/5 (mouse #4); 20/5, and 17/4 (mouse #5), respectively. For Figs. 2–5 & 7–8, all recorded neurons (n = 376) from the 15 sessions were included in the analyses. For Fig. 6, however, one or two sessions were excluded because they contained too few trials of certain types (n < 3).

The discrimination–avoidance task was similar to that during the training phase, except that five direct-current electric shocks, which minimize electromagnetic artifacts, were administered starting at sound terminations and continued for 8 s. This shorter shock period allowed the mice more time to move freely within the shuttle box during the 42-s inter-trial intervals, when no consequences were administered. Additionally, each task session included a 5-min free exploration period before trials began. In some sessions, a small number of trials were excluded from analysis, partly due to the electrode implant or recording cable occasionally interfering with the animal's shuttle responses.

Shuttle behavior analysis

We used DeepLabCut [60] to analyze animals' locations during discrimination–avoidance tasks, with all locations determined based on body center positions. Shuttle crossing was defined as the body center crossing the midline opening of the shuttle box. Shuttle initiations and terminations were defined as the time points when the animal's movement velocity deviated 1 s.d. above mean (Fig. 4A).

Dimension reduction

We used Principal Component Analysis (PCA) to analyze the major activity patterns of ACC neuronal populations. The first three principal components (PCs) were used for 3-D visualizations. Specifically, the activity of each ACC neuron was first averaged during shuttle responses (bin size, 0.1 s) and z scored. The z-scored activity of ACC neurons was then processed with PCA (*pca*, MATLAB).

Speed tuning analysis and speed cell classification

Speed tuning was assessed using neural and behavioral data collected during a ~5-min period of free movement in the shuttle box prior to task onset. During this period, animals freely explored the environment and exhibited a broad range of instantaneous running speeds. Running speed was extracted from behavioral tracking data and sampled at 50 Hz. Spike trains from individual neurons were binned at 20 ms resolution and converted to firing rates. Firing-rate time series were smoothed using a Gaussian kernel with a 250 ms window. To reduce potential confounds associated with immobility-related network states, time points with running speed below 2 cm/s were excluded from the analysis, following established procedures [61]. For each neuron, a speed score was defined as the Pearson correlation coefficient between the smoothed firing rate and the valid portion of the instantaneous running speed.

Statistical significance was assessed using a circular time-shift shuffle procedure. For each neuron, spike times were circularly shifted by a random offset greater than 30 s, preserving the temporal structure of firing while disrupting its alignment with behavior. The firing rate–speed correlation was recomputed for each shuffle, and this procedure was repeated 100 times to generate a null distribution of speed scores. Neurons were classified as speed-modulated if their observed speed score exceeded the 99th percentile or fell below the 1st percentile of the shuffle distribution (two-sided criterion). All analysis procedures were adapted from previous studies of speed-modulated neurons [61].

PCA-based classification of ACC neuronal types

We used PCA to classify major types of ACC neurons based on their activity in reference to shuttle initiation. Specifically, the activity of each ACC neuron was first averaged surrounding shuttle responses (−5–5 s; bin size, 25 ms) and z-scored. The first three PCs were used in a hierarchical clustering algorithm (Linkage) to find the similarity (Euclidean distance) between all pairs of activity patterns in PC space, iteratively grouping the activity patterns into larger and larger clusters based on their similarity. Lastly, we set a distance-criterion to extract major clusters from the hierarchical tree [62].

GLM-based classification of 'shuttle-content' and 'shuttle-state' neurons

For each neuron, we computed peri-event firing rates using 25-ms bins and normalized activity relative to a baseline window from -30 to -20 s preceding shuttle onset. Post-shuttle activity (0–5 s) was analyzed using a generalized linear model (GLM) with separate regressors for each event. Specifically, for Eventl (shuttle A→B) and Event2 (shuttle B→A) trials, we fit the model

$$r(t) = \beta 1 \cdot \mathrm{Event1}(t) + \beta 2 \cdot \mathrm{Event2}(t),$$

using a normal distribution with an identity link (*glmfit*, constant = 'off', MATLAB). We estimated coefficients β1 and β2 (corresponding to shuttle A→B and shuttle B→A, respectively) and assessed their significance using Wald tests against zero. To quantify event selectivity, we tested the contrast: $\Delta\beta = \beta 1 - \beta 2$. Resulting p-values were corrected for multiple comparisons using Bonferroni correction across regressors and neurons ($\alpha\_bonf = 0.01/(3N)$, where N is the number of neurons in a session). Neurons were classified as *action-content* neurons if the corrected p-value for Δβ was significant and the absolute effect size exceeded a predefined threshold ($|\Delta\beta| > 0.5$). Neurons were classified as *action-state* neurons if Δβ was not significant but both β1 and β2 were individually significant after correction. All analysis procedures were adapted from previous studies [63–65].

Machine learning decoding

We used Support Vector Machine (SVM) classifiers to train ACC neuronal population activity to decode shuttle content (rooms A→B *ys.* B→A shuttles) and subsequently performed 10-fold cross-validations. Specifically, the total number of spikes from each ACC neuron, calculated during either the pre-shuttle (−5–0 s) or post-shuttle period (0–5 s), were used for training and testing the SVM classifier. For each dataset, we performed SVM classification training and cross-validation 100 times (*fitcsym* and *crossyal*, MATLAB), assigning the mean correct classification rate as the decoding accuracy. Notably, using shorter or longer time windows surrounding shuttle responses, such as 2.5 or 10 s, yielded similar conclusions (Fig. S7). Additionally, ACC neuronal population activity moderately decoded Room-A *ys.* Room-B stays (Fig. S7; SVM training and cross-validation repeated 500 times).

To assess the contribution of *action-content* neurons to decoding performance, neurons were progressively removed in 20% increments (Fig. 8C). Within each session, *action-content* neurons were ranked by | Δβ | in descending order, and the top fraction was removed at each step prior to re-estimating decoding accuracy. As a control, the same number of neurons was randomly removed from the Δβ non-significant pool. At each removal fraction, decoding accuracies obtained after *action-content* neuron removal were compared to control accuracies across sessions using a paired, two-sided, Wilcoxon signed-rank test.

Spatial information analysis (Fig. S6)

To quantify spatial information coding, firing rate maps for individual neurons were constructed using 1 × 1 cm spatial bins in NeuroExplorer and smoothed with a Gaussian filter (filter width, 3 bins) before being exported to MATLAB for further analyses. Spatial information for each neuron was calculated using the following formula:

$$Spatial\ Information\ (bits/spike) = \sum_{i} p_i \frac{\lambda_i}{\lambda} \log_2 \frac{\lambda_i}{\lambda},$$

where λi is the mean firing rate in the i-th bin, λ is the overall mean firing rate, and p, is the probability of the animal's being in the i-th bin, as described previously [55, 66].

Histology

To mark the final recording sites, we made electrical lesions by passing 10-s, 10-μA currents through multiple tetrodes. Mice were deeply anesthetized and intracardially perfused with ice-cold PBS or saline, followed by 10% formalin. The brains were removed and postfixed in formalin for at least 24 hours. The brains were sliced into coronal sections of 50-μm thickness using *Leica* vibratome. Brain sections were mounted with Mowiol mounting medium for microscopic examination of electrode array placements.

Statistics

Sample sizes were based on previous similar studies [55, 62]. All statistics were conducted in SPSS 30.0. Statistical analyses include repeated measures ANOVA, the nonparametric Friedman test by post-hoc test (pair-wise Wilcoxon signed-rank test and Bonferroni correction), the Wilcoxon signed-rank test (paired), and Student's *t* test (paired). All statistical tests are two-sided; P-values of 0.05 or lower were considered significant.

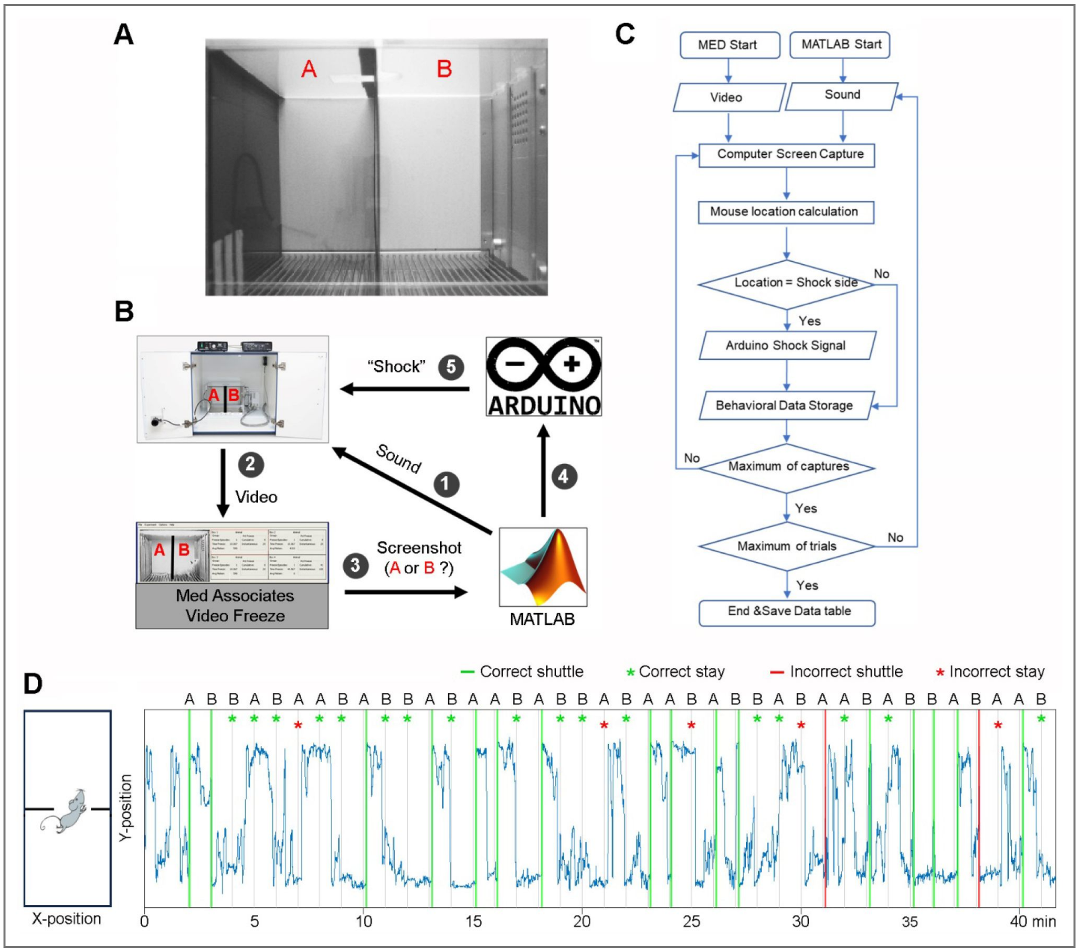


Fig. S1. **Experimental setup. A**, The shuttle box used for behavioral training. Room A is configured with two black walls (left and right), one white wall (back), and a transparent front wall for video recording purposes. Room B is configured with two white walls (left and back), one metal wall (right), and a transparent front wall. **B**, Schematic diagram of the control setup utilizing MATLAB functions, with numbers indicating the sequence of control flow. **C**, A comprehensive flowchart illustrating the control setup as shown in B. **D**, Left, sounds A and B signal shocks in the bottom and top rooms of the shuttle box, respectively. Right, the Y position of a well-trained mouse in a ∼40-min session.

Fig. S2. Extended training recording sites.

A, Correct and incorrect trial counts for shuttle *vs.* stay trials across 15 training sessions. In shuttle trials, mice must move to the adjacent room before the sound ends to avoid shocks, whereas in stay trials, mice must remain in their current room to avoid shocks. **B**, reconstructed recording sites for individual mice. **C**, Heatmaps showing the activity of all recorded ACC neurons during correct-shuttle trials across individual mice.

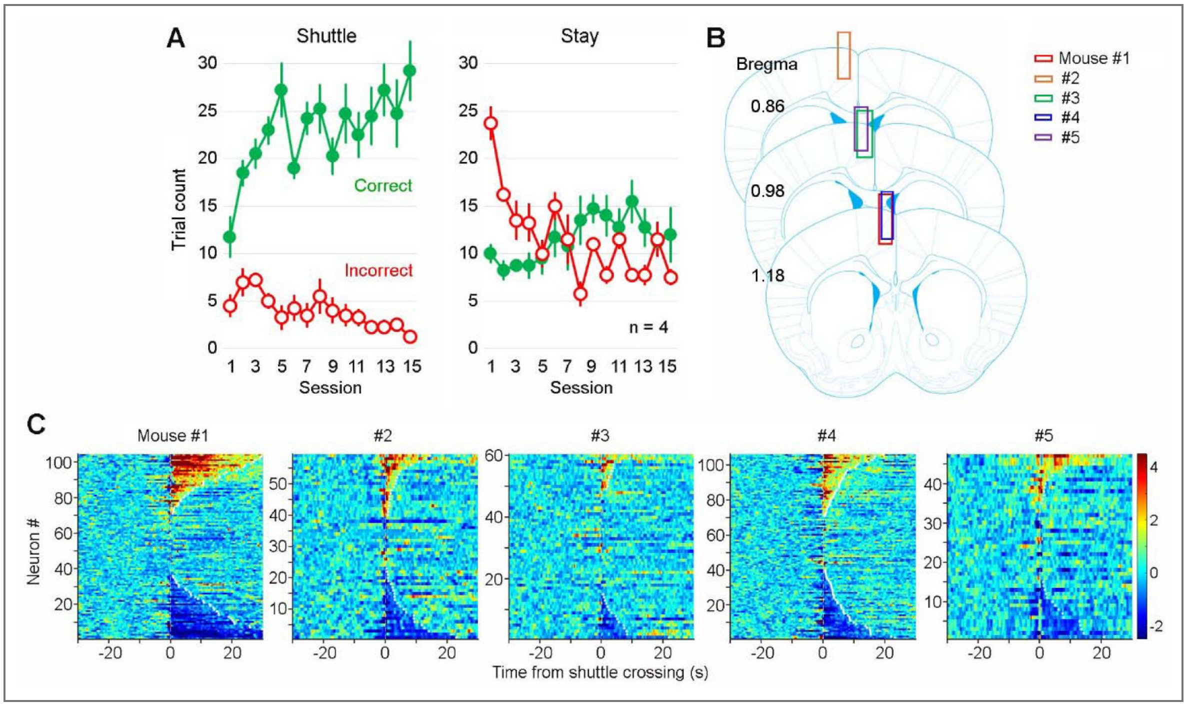


Fig. S3. ACC neurons primarily respond during "shuttle" but not "stay" trials.

A&B, Dimension reduction analysis (i.e., PCA) indicates robust changes in ACC neuronal population activity during correct shuttles (A) but not correct stays (B) from four representative recording sessions. Time "0" indicates shuttle crossings (A) or sound onsets (B), respectively. For more details, see Fig. 3B.

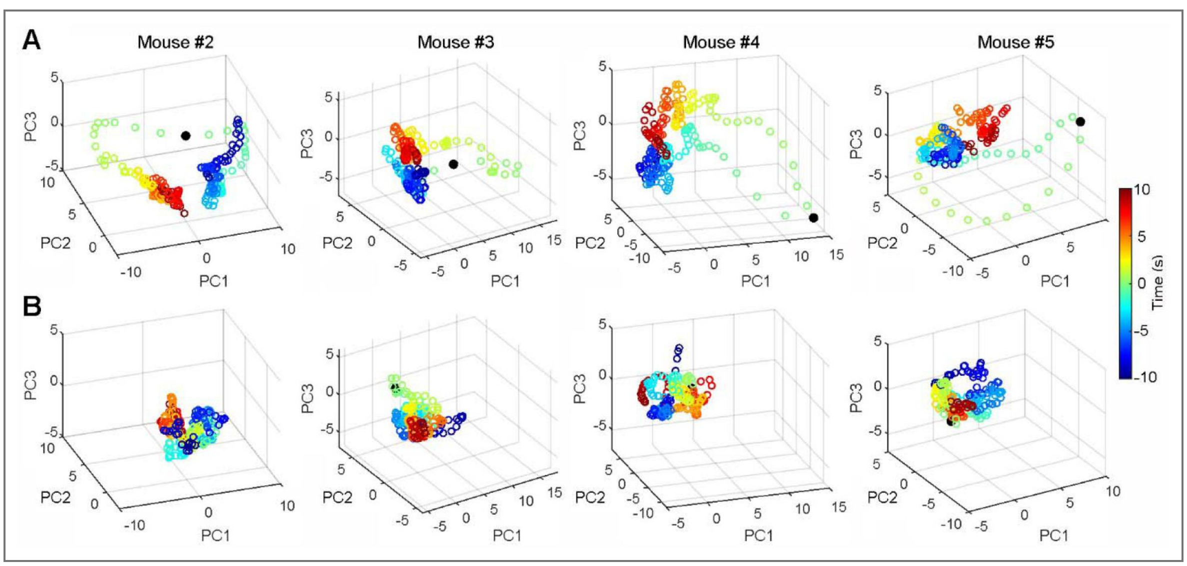


Fig. S4. **ACC shows limited response to cue, stay trials, and footshocks.**

Left, Heatmap showing the activity of individual ACC neurons (n = 348) in relation to auditory cue onset. **Middle**, Heatmap showi**ng** the activity of individual ACC neurons (n = 348) during stay trials. **C**, Heatmap showing the activity **of** individual ACC neurons (n = 336) during footshock. Note, footshock trials with a shuttle response within **1** s shock onset were excluded to avoid shuttle response confound. As a result, some behavioral sessio**n**s were excluded due to insufficient trial numbers.

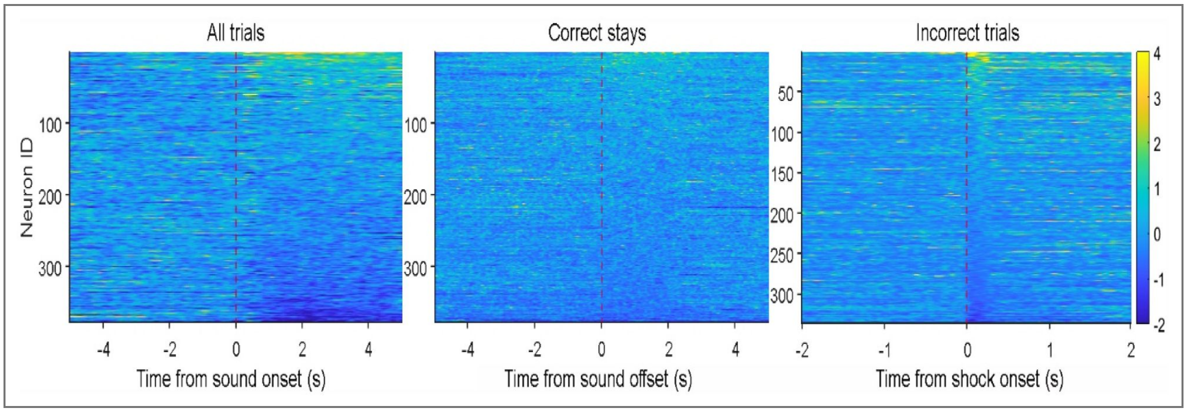


Fig. S5. **ACC pyramidal neurons and interneurons both monitor *action content*.**

A, Spike waveforms (mean ± s.d.) of two representative ACC neurons: one putative pyramidal neuron and one interneuron. **B**, Peri-event rasters (trials) & histograms of the same two ACC neurons surrounding shuttle response**s.** Both neurons exhibit differential activity changes that discriminate between rooms A B (top panels) ***vs.*** B A shuttles (bottom panels). **C**, Cross-correlation histograms between the putative interneuron (Neuro**n** #2; the same as shown in A) and three other pyramidal neurons (Neurons #1, #3, and #4). Neuron #**4** appears to excite Neuron #2, which in turn inhibits Neuron #3, as indicated by short-latency (~2 m**s)** excitatory or inhibitory interactions. The four ACC neurons were recorded simultaneously.

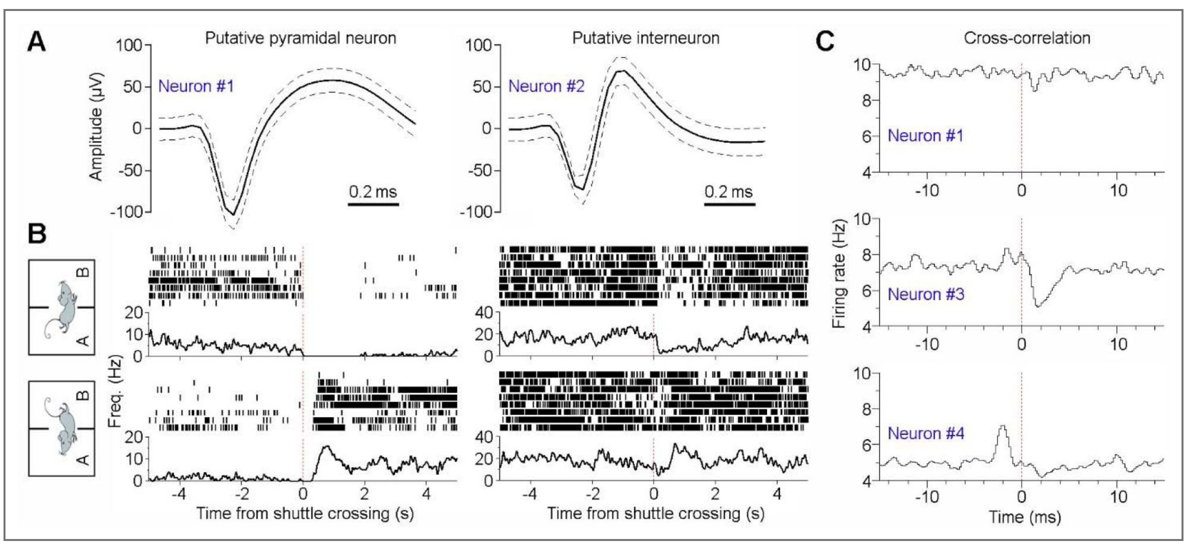


Fig. S6. **ACC neurons do not display place cell characteristics.**

A, A representative navigation path during inter-trial intervals of a discrimination-avoidance task session. **B**, Place field activity of two representative ACC neurons. Both neurons show spiking activity across the chamber without place preference. The color bar indicates normalized firing rate. **C**, Spatial information coding distribution of all recorded ACC neurons (n = 376). **D–F**, Same as in A–C, but for neurons recorded from hippocampal dorsal CA1 (n = 161). Notably, more than half of CA1 neurons exhibit high information coding (>2 bits/spike; F), whereas very few ACC neurons show comparable levels of information coding (C).

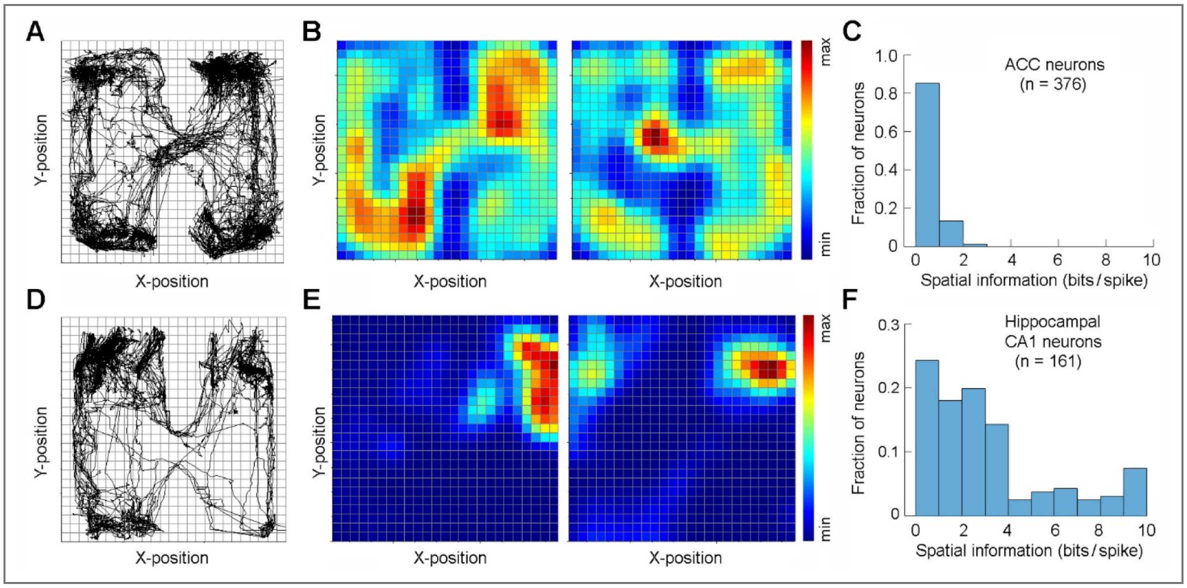
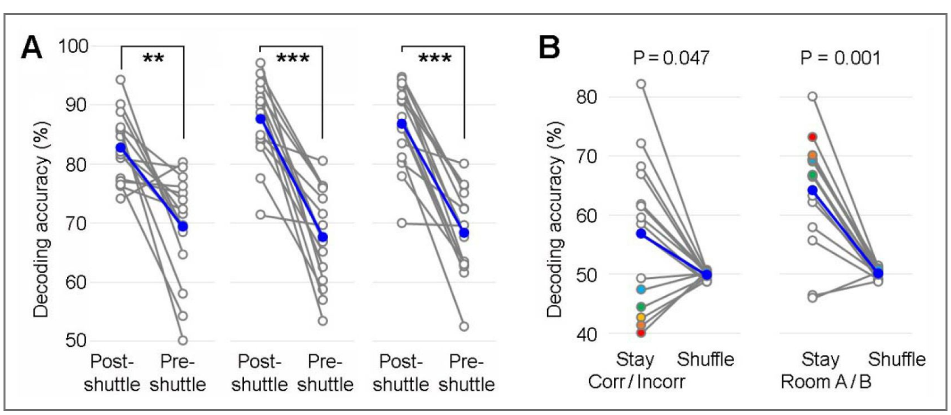


Fig. S7. **SVM decoding of multiple behaviors from ACC neuronal population activity.**

A, Mean (blue line) and individual session decoding accuracies (grey lines; 15 sessions) for decoding A B *vs.* B **A** shuttles using short (left, 2.5 s), medium (middle, 7.5 s), or long decoding windows (right, 10 s) surrounding shuttle crossings. **P < 0.01, ***P < 0.001, *Wilcoxon signed-rank test*. **B**, Mean (blue line) and individual session decoding accuracies (grey lines) for decoding correct *vs.* incorrect stays (left) or Room-A *vs.* Room-B stays (right), using a 10-s window after sound onset. Notably, in several sessions, lower-than-chance decoding accuracy for Corr/Incorr stays (colored dots; left) corresponded to high decoding accuracy for Room A/B stays (shown in the same colors; right), suggesting a potential confound between these variables.



Data availability

All essential behavioral and electrophysiological data will be publicly available to accompany the publication of the final version of this paper. The code for the analyses presented in this paper will also be openly accessible.

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Additional files

Supplemental Video 1

Supplemental Video 2

Additional information

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Peer reviews

Reviewer #1 (Public review):

Summary:

Huang et al. examined ACC response during a novel discrimination-avoid task. The authors concluded that ACC neurons primarily encode post-action variables over extended periods, reflecting the animal's preceding actions rather than the outcomes or values of those actions. The authors have made considerable revisions to address the raised concerns. However, it appears that some important issues remain unresolved.

Strengths:

The inclusion of new figures and analyses in response to the reviews is appreciated, such as Fig. 2 and 5.

Weaknesses:

Motion related signal in ACC: the new Fig. 2E looks good, but it is hard to visualize how it is just a reordering of the old Fig. 5C.

All categories in the new Fig. 4D appear to respond to shuttle initiation, with less than 1s latency. For example, type 2a/2b consists of 40% of the population and their response to movement onset is apparent. Thus, it is not clear whether most neurons respond to shuttle crossing as described in the manuscript.

Could the authors use relatively simple analysis, such as comparing spike rate before and after crossing, or before and after initiation, to quantify the response properties of each neuron? This could also help validate the classification analysis performed in Fig. 4.

https://doi.org/10.7554/eLife.105774.2.sa3

Reviewer #2 (Public review):

Summary:

Huang et al recorded anterior cingulate cortex activity in mice while they performed a shuttle escape task. The task utilized two auditory cues, each of which informed the mice to stay or escape depending on which side they were on, and incorrect responses were punished by shock administration. Analyses focused on ACC neurons that fired when mice crossed the shuttle box in either direction (A-->B or B-->A), coined "action state", or when mice crossed in one direction but not the other, coined "action content". The authors characterized these populations, and ACC firing changes mostly occurred around the time of shuttle crossing. This work will likely be of broad interest to those who are interested in neocortical neurophysiology broadly, anterior cingulate cortex specifically, and their contributions to learning about actions. The task is well-designed and provides a nice background for neurophysiological recordings. The authors leveraged these strengths in characterizing the neural populations that fire to shuttle crossings in both directions vs one direction.

Strengths:

The factorial design nicely controls for sensory coding and value coding, since the same stimulus can signal different actions and values.

The figures are well presented, labeled, and easy to read.

Additional analyses, such as the 2.5/7.5s windows and place-field analysis, are nice to see and indicate that the authors were careful in their neural analyses.

The n-trial + 1 analysis where ACC activity was higher on trials that preceded correct responses is a nice addition, since it shows that ACC activity predicts future behavior, well before it happens.

The authors identified ACC neurons that fire to shuttle crossings in one direction or to crossings in both directions. This is very clear in the spike rasters and population scaled color images. While other factors such as place fields, sensory input, and their integration can account for this activity, the authors discuss this and provide additional supplemental analyses.

Weaknesses:

Some of the neural analyses could use the necessary and sufficient comparisons to strengthen the authors' claims.

Comment on revised version:

I think the authors did a very admirable job revising the manuscript. It is much improved. However, I believe a formal analysis of action-state versus action-content neurons on A-->B versus B-->A crossing is still warranted. I appreciate the fact that this analysis may not be as reliable with smaller ensemble sizes, but with careful pseudo-ensemble and resampling approaches, such an analysis would go a long way towards increasing the strength of evidence.

https://doi.org/10.7554/eLife.105774.2.sa2

Reviewer #3 (Public review):

Summary:

The authors record from the ACC during a task in which animals must switch contexts to avoid shock as instructed by a cue. As expected, they find neurons that encode context, with some encoding of actions prior to the context, and encoding of neurons post-action. The primary novelty is dynamic encoding of action-outcome in a discrimination-avoidance domain, while this is traditionally done using operant methods.

Comments on revised version:

I appreciate the considerable work done on review, and additional details added throughout. I also noted the additional sessions included in analyses, and additional behavioral data in response to R1 and R2's insightful comments.

The only remaining comment that was not addressed pertains to anatomy and recording details. Some electrodes appear to be clearly in M2 (Fig 2A), and the tetrodes were driven each day. I would strongly suggest that this be included as a further limitation, particularly given the statement on line 178.

https://doi.org/10.7554/eLife.105774.2.sa1

Author response:

The following is the authors' response to the original reviews.

> ***Public Reviews:***
>
> ***Reviewer #1 (Public review):***
>
> *Summary:*
>
> *In the current study, Huang et al. examined ACC response during a novel discrimination-avoid task. The authors concluded that ACC neurons primarily encode post-action variables over extended periods, reflecting the animal's preceding actions rather than the outcomes or values of those actions. Specifically, they identified two subgroups of ACC neurons that responded to different aspects of the actions. This work represents admirable efforts to investigate the role of ACC in task-performing mice. However, in my opinion, alternative explanations of the data were not sufficiently explored, and some key findings were not well supported.*
>
> *Strengths:*
>
> *The development of the new discrimination-avoid task is applauded. Single-unit electrophysiology in task-performing animals represents admirable efforts and the datasets are valuable. The identification of different groups of encoding neurons in ACC can be potentially important.*
>
> *Weaknesses:*
>
> *One major conclusion is that ACC primarily encodes the so-called post-action variables (specifically shuttle crossing). However, only a single example session was included in Figure 2, while in Supplementary Figure 2 a considerable fraction of ACC neurons appears to respond to either the onset of movement or ramp up their activity prior to movement onset. How did the authors reach the conclusion that ACC preferentially respond to shuttle crossing?*

We now include more example sessions and the main results from individual animals (Fig. 3; Figs. S2–S3; Fig. 8). Overall, the results are consistent across recording sessions and animals.

While shuttle crossings were the primary reference for most analysis, using shuttle initiation as a reference led to similar conclusions (Fig.4). Namely, we found that most ACC neurons exhibit either robust (22%; Types 1a & 2a) or moderate (51%; Types 1b & 2b) post-shuttle activity changes (Fig.4), while only a subset exhibits ramping pre-shuttle activity (16%; Types 3b & 3c). Therefore, our conclusion was intended to highlight the role of post-shuttle activity in learning. While we do not exclude the possibility that pre-shuttle ACC activity contributes to learning, its involvement is likely more limited.

> *In Figure 4, it was concluded that ACC neurons respond to action independent of outcome. Since these neurons are active on both correct and incorrect shuttle but not stay trials, they seem to primarily respond to overt movement. If so, the rationale for linking ACC activity and adaptive behavior/ associative learning is not very clear to me. Further analyses are needed to test whether their firing rates correlated with locomotion speed or acceleration/deceleration. On a similar note, to what extent are the action state neurons actually responding to locomotion-related signals? And can ACC activity actually differentiate correct vs. incorrect stays?*

In this study, we highlight two distinct groups of ACC neurons: action-state and action-content neurons. Both groups of neurons tend to show sustained activity even when the animals remain immobile after completing shuttle behaviors, suggesting that their activity is not directly driven by locomotion. Furthermore, action-content neurons are selectively engaged in only one of the two shuttle categories, either rooms A→B or B→A shuttles. Therefore, differences in neuronal activity are unlikely to reflect locomotor differences, given that both shuttle types involve similar movement patterns. Finally, we analyzed ACC neuronal activity in relation to locomotion speed. Our results indicate that only a small fraction of neurons (<15%) show speed-correlated activity (Fig.5), suggesting that most ACC neurons do not encode movement-related information. Taken together, these findings support the distinction between ACC activity and locomotion encoding.

As for the small subset of speed-related neurons, it remains unclear whether these speed-related neurons represent a distinct subpopulation within the ACC or reflect recordings from the nearby motor cortex. Postmortem examination of the recording sites suggests that most neurons were recorded from the ACC, while a small subset may be located at the border between the ACC and motor cortex (Fig. S2). Therefore, it is possible that the small fraction of speed-related neurons originated from the motor cortex.

Lastly, given that the ACC neurons display no or limited activity during stay trials, their activity generally does not differentiate correct vs. incorrect stays (Fig.S7). However, ACC activity does show moderate differentiation between room-A vs. room-B stays (Fig.S7).

> *Given that a considerable amount of ACC neurons encode 'action content', it is not surprising that by including all neurons the model is able to make accurate predictions in Figure 6. How would the model performance change by removing the content neurons?*

We thank the reviewer for this thoughtful analysis idea. Excluding action-content neurons drastically reduces decoding accuracy (Fig.8), suggesting that they are the main drivers for differentiating rooms A→B vs. B→A shuttles.

> *Moving on to Figure 7. Since Figure 4 showed that ACC neurons respond to movement regardless of outcome, it is somewhat puzzling how ACC activity can be linked to future performance.*

As discussed earlier (point #2), ACC activity does not simply reflect locomotion itself. We interpret the post-shuttle ACC activity as encoding both the preceding shuttle state (shuttle or stay) and shuttle content (rooms A→B or B→A). Regardless of the outcome (safety or shock), such encoding is essential for cue–action–outcome associative learning, because both positive and negative feedback can drive learning. The level of post-shuttle ACC activity may reflect task engagement, with greater engagement facilitating learning and improving future performance.

> *Two mice contributed about 50% of all the recorded cells. How robust are the results when analyzing mouse by mouse?*

We have added further analysis of highlighting the results of each mouse. Although the total number of recorded neurons varied across mice, the major findings were consistent. In every mouse, we observed sustained post-shuttle ACC activity (Fig.S2), and population-level ACC activity reliably decoded shuttle contents (rooms A→B vs. B→A; Fig.8).

> *Lastly, the development of the new discrimination-avoid task is applauded. However, a major missing piece here is to show the importance of ACC in this task and what aspects of this behavior require ACC.*

We appreciate this feedback. We are currently conducting additional experiments to determine whether inhibiting ACC activity during distinct time windows disrupts task learning. We hope to publish a follow-up paper on these findings in the near future.

> ***Reviewer #2 (Public review):***
>
> *Summary:*
>
> *The current dataset utilized a 2x2 factorial shuttle-escape task in combination with extracellular single-unit recording in the anterior cingulate cortex (ACC) of mice to determine ACC action coding. The contributions of neocortical signaling to action-outcome learning as assessed by behavioral tasks outside of the prototypical reward versus non-reward or punished vs non-punished is an important and relevant research topic, given that ACC plays a clear role in several human neurological and psychiatric conditions. The authors present useful findings regarding the role of ACC in action monitoring and learning. The core methods themselves - electrophysiology and behavior - are adequate; however, the analyses are incomplete since ruling out alternative explanations for neural activity, such as movement itself, requires substantial control analyses, and details on statistical methods are not clear.*
>
> *Strengths:*
>
> *(1) The factorial design nicely controls for sensory coding and value coding, since the same stimulus can signal different actions and values.*
>
> *(2) The figures are mostly well-presented, labeled, and easy to read.*
>
> *(3) Additional analyses, such as the 2.5/7.5s windows and place-field analysis, are nice to see and indicate that the authors were careful in their neural analyses.*
>
> *(4) The n-trial + 1 analysis where ACC activity was higher on trials that preceded correct responses is a nice addition, since it shows that ACC activity predicts future behavior, well before it happens.*
>
> *(5) The authors identified ACC neurons that fire to shuttle crossings in one direction or to crossings in both directions. This is very clear in the spike rasters and population-scaled color images. While other factors such as place fields, sensory input, and their*

integration can account for this activity, the authors discuss this and provide additional supplemental analyses.

Weaknesses:

(1) The behavioral data could use slightly more characterization, such as separating stay versus shuttle trials.

We appreciate this feedback. In the revised manuscript, we present data separating stay versus shuttle trials (Fig.1). Additionally, we provide new data from extended training sessions (Fig.S2).

(2) Some of the neural analyses could use the necessary and sufficient comparisons to strengthen the authors' claims.

We have now used the necessary and sufficient comparisons where applicable. In the SVM decoding analysis, we show that population ACC activity is sufficient to decode A→B or B→A shuttles. We also show that excluding action-content, but not other ACC neurons, drastically reduces decoding accuracy, suggesting that these neurons are necessary for the decoding (Fig.8).

(3) Many of the neural analyses seem to utilize long time windows, not leveraging the very real strength of recording spike times. Specifics on the exact neural activity binning/averaging, tests, classifier validation, and methods for quantification are difficult to find.

We chose to perform our neural analyses on a longer time scale, given the sustained activity we see in the data. To further justify that decision, we now provide additional results highlighting the sustained activity of ACC neurons in our task (Fig.2; Fig.S2). Additionally, we now provide more specifics of the neural analyses in Methods section.

(4) The neural analyses seem to suggest that ACC neurons encode one variable or the other, but are there any that multiplex? Given the overwhelming evidence of multiplexing in the ACC a bit more discussion of its presence or absence is warranted.

This is an interesting point of discussion, and we thank the reviewer for pointing this out. Overall, our results suggest that individual ACC neurons preferentially engage in only one of the proposed functions, rather than multiplexing across them. For example, action-state and action-content ACC neurons primarily engage in action monitoring, but not in decision-making, planning, or outcome tracking. Nevertheless, we cannot rule out the possibility that other ACC neurons, through their distinct connectivity or location in different ACC subregions, engage in other proposed functions. Thus, when considering the ACC as a whole, its function may still be multiplexed.

Another possible reason we do not see clear multiplexing of neurons may be due to the dynamic nature of our task. Unlike established tasks that often assign fixed positive or negative values to cues, the cues in our task are not inherently associated with valence. Instead, their meaning is dynamically determined by the animal's location (context) at the time of cue presentation. Since values are not fixed and change based on context, value-related responses may not be reflected in the ACC in our tasks.

We have now incorporated the above discussions into our revised manuscript.

Reviewer #3 (Public review):

Summary:

> *The authors record from the ACC during a task in which animals must switch contexts to avoid shock as instructed by a cue. As expected, they find neurons that encode context, with some encoding of actions prior to the context, and encoding of neurons post-action. The primary novelty of the task seems to be dynamically encoding action-outcome in a discrimination-avoidance domain, while this is traditionally done using operant methods. While I'm not sure that this task is all that novel, I can't recall this being applied to the frontal cortex before, and this extends the well-known action/context/post-context encoding of ACC to the discrimination-avoidance domain.*
>
> *While the analysis is well done, there are several points that I believe should be elaborated upon. First, I had questions about several details (see point 3 below). Second, I wonder why the authors downplayed the clear action coding of ACC ensembles. Third, I wonder if the purported 'novelty' of the task (which I'm not sure of) and pseudo-debate on ACC's role undermines the real novelty - action/context/outcome encoding of ACC in discrimination-avoidance and early learning.*
>
> *Strengths:*
>
> *Recording frontal cortical ensembles during this task is particularly novel, and the analyses are sophisticated. The task has the potential to generate elegant comparisons of action and outcome, and the analyses are sophisticated.*
>
> *Weaknesses:*
>
> *I had some questions that might help me understand this work better.*
>
> *(1) I wonder if the field would agree that there is a true 'debate' and 'controversy' about the ACC and conflict monitoring, or if this is a pseudodebate (Line 34). They cite 2 very old papers to support this point. I might reframe this in terms of the frontal cortex studying action-outcome associations in discrimination-avoidance, as the bulk of evidence in rodents comes from overtrained operant behavior, and in humans comes from high-level tasks, and humans are unlikely to get aversive stimuli such as shocks.*

We appreciate this feedback. We have revised the Introduction and Discussion.

> *(2) Does the purported novelty of the task undermine the argument? While I don't have an exhaustive knowledge of this behavior, the novelty involves applying this ACC. There are many paradigms where a shock triggers some action that could be antecedents to this task.*

We argue our newly designed discrimination–avoidance task is unique for several reasons. First, it requires animals to discriminate both sensory cues and environment contexts. Unlike established tasks that often assign fixed positive or negative values to cues, the cues in our task are not inherently associated with valence. Instead, their meaning is dynamically determined by the animal’s location (context) at the time of cue presentation, which reflects a conceptual advance over previous techniques. Furthermore, by removing valence from the cues, this design helps disentangle the ACC’s potential role in value encoding from other cognitive functions.

Second, this task involves robust, ethologically relevant actions (i.e., shuttles), unlike many established paradigms that rely on less naturalistic behaviors such as saccades or lever presses. We view this as a key distinction from prior approaches, as even previous paradigms that utilize shutting responses or other naturalistic responses, fail to incorporate dynamic integration of cues and contexts.

Finally, the clear temporal separation between actions and outcomes further helps disentangle the ACC's roles in action monitoring vs. outcome tracking.

> *(3) The lack of details was confusing to me:*
>
> *(a) How many total mice? Are the same mice in all analyses? Are the same neurons? Which training day? Is it 4 mice in Figure 3? Five mice in line 382? An accounting of mice should be in the methods. All data points and figures should have the number of neurons and mice clearly indicated, along with a table. Without these details, it is challenging to interpret the findings.*

We are sorry for the confusion. We now provide additional details and clear N numbers for each analysis to improve clarity.

> *(b) How many neurons are from which stage of training? In some figures, I see 325, in some ~350, and in S5/S2B, 370. The number of neurons should be clearly indicated in each figure, and perhaps a table.*

All data were obtained from well-trained mice. For some analyses, the N is smaller because certain task sessions contained very few incorrect trials (≤3), which prevented us from examining ACC activity during those trials. We have modified figure legend so that neuron count is clear.

> *(c) Were the tetrodes driven deeper each day? The depth should be used as a regressor in all analyses?*

Yes, the tetrodes were driven slightly deeper across task sessions (~80 µm per step; 2–4 depths per mouse). Given limited depth changes, preliminary analyses indicate no clear differences in ACC activity across these recording depths. However, we cannot rule out potential dorsal–ventral subregion differences if recordings were to span larger depth ranges.

> *(d) Was is really ACC (Figure 2A)? Some shanks are in M2? All electrodes from all mice need to be plotted as a main figure with the drive length indicated.*

We have now included a supplementary figure showing all recording sites (Fig.S2). It is likely that a small subset of neurons was recorded at the ACC/M2 border area. Unfortunately, we are unable to separate them out due to blind recording design of our tetrode arrays.

> *(e) It's not clear which sessions and how many go into which analysis*

We have now specified the number of task sessions for each analysis (see Methods).

> *(f) How many correct and incorrect trials (<7?) are there per session?*

We have now specified the number of correct and incorrect trials per session (see Methods).

> *(g) Why 'up to 10 shocks' on line 358? What amplitudes were tried? What does scrambled mean?*

We decided to use up to 10 mild shocks per trial because mice do not necessarily shuttle to the safe room after one or even a few shocks during the early stages of training. This design allows mice to efficiently learn the concept of the task (i.e., one room is safe while the other delivers shocks). Each shock was specified in the Methods section as 0.5 mA, 0.1 s. A "scrambled shock" refers to an electric shock delivered through multiple floor bars in a randomized pattern, effectively preventing the animal from avoiding the stimulus.

> *(4) Why do the authors downplay pre-action encoding? It is clearly evident in the PETHs, and the classifiers are above chance. It's not surprising that post-shuttle classification is*

> *so high because the behavior has occurred. This is most evident in Figure S2B, which likely should be a main figure.*

We did not intend to downplay pre-action encoding. Our analysis shows that most ACC neurons exhibit either robust (22%; Types 1a & 2a) or moderate (51%;Types 1b & 2b) post-shuttle activity changes (Fig.4). Although a subset of ACC neurons exhibits ramping pre-shuttle activity, they represent a much smaller fraction (16%; Types 3b & 3c). Therefore, our conclusion was intended to highlight the role of post-shuttle activity in learning. While we do not exclude the possibility that pre-shuttle ACC activity contributes to learning, its involvement is likely more limited

> *(5) The statistics seem inappropriate. A linear mixed effects model accounting for between-mouse variance seems most appropriate. Statistical power or effect size is needed to interpret these results. This is important in analyses like Figure 7C or 6B.*

We appreciate this feedback. We now use appropriate statistics and report effect size.

> *(6) Better behavioral details might help readers understand the task. These can be pulled from Figures S2 and S5. This is particularly important in a 'novel' task.*

We now provide more details to help better understand the task and have added new figures (Fig.1; Figs. S1&S2).

> *(7) Can the authors put post-action encoding on the same classification accuracy axes as Figure 6B? It'd be useful to compare.*

We appreciate the comment, but we are unsure what clarification is being requested.

> *(8) What limitations are there? I can think of several - number of animals, lack of causal manipulations, ACC in rodents and humans.*

We now include discussions on limitation of our study. One caveat of our study is that the discrimination–avoidance task requires weeks of training in mice. By the time they master the task, ACC activity may reflect modified neural circuits. Investigating ACC activity during early phase of learning, such as by introducing a new pair of cues or contexts, could provide further insights into ACC's role in learning and cognitive processes. Additionally, a limitation of the current study is the lack of evidence for the causal role of post-action ACC activity in complex associative learning. Future investigations using closed-loop strategies to selectively disrupt ACC activity during the post-action phase could help address this question.

> *Minor:*
>
> *(1) Each PCA analysis needs a scree plot to understand the variance explained.*

We have added a scree plot for each PCA analysis.

> *(2) Figure 4C - y and x-axes have the same label?*

We have corrected the y-axis label.

> *(3) What bin size do the authors use for machine learning (Not clear from line 416)?*

The bin sizes used were 2.5, 5, 7.5, or 10 sec which have now been discussed in the Methods section.

> *(4) Why not just use PCA instead of 'dimension reduction' (of which there are many?)*

We have adjusted the phrasing where appropriate.

> *(5) Would a video enhance understanding of the behavior?*

We appreciate this feedback. We now include a few videos to accompany our paper.

> ***Recommendations for the authors:***
>
> ***Reviewer #1 (Recommendations for the authors):***
>
> *(1) Is Figure 1C sufficiently powered?*

We have now included data from additional mice and updated the figure accordingly.

> *(2) Task performance was not plateaued after 10 sessions in Figure 1B. How variable is task performance in the datasets with ephys recordings (session to session, mouse to mouse).*

We have now included additional data from extended training (15 sessions; Fig.S2). Moderate variations across both sessions and mice are observed. Specifically, the total number of correct/incorrect shuttles used for ephys analysis are 19/5, 19/4, 21/5, 20/4 (mouse #1; 4 sessions); 20/7, 23/7, 20/7 (mouse #2; 3 sessions); 19/4, 16/2 (mouse #3; 2 sessions); 26/4, 23/4, 17/6, 25/5 (mouse #4; 4 sessions); 20/5, and 17/4 (mouse #5; 2 sessions), respectively.

> *(3) Please quantify the results in Figure 3, for both within individual mice and across mice.*

We have calculated maximum trajectory length within the 3-D space (Fig. 3C).

> *(4) What is the effect size in Figure 7C?*

We now report the effect size.

> *(5) Please provide more details for spike sorting.*

We have now included more details in the Methods section.

> *(6) More detailed cell type or correlation analysis in Figures 4 and 5 may be helpful. For example, if putative regular and fast-spiking neurons were simultaneously recorded, did the FS directly inhibit the RS to give rise to the apparent encoding properties?*

We recorded a small number of putative interneurons (n = 13) from only three mice, which precludes drawing meaningful conclusions, particularly given their heterogeneous responses during discrimination–avoidance tasks. Accordingly, we include only an example interneuron demonstrating discrimination between A→B vs. B→A shuttles (Fig. S5). Nevertheless, it is evident there are reciprocal monosynaptic connections between putative interneurons and certain pyramidal neurons, as indicated by short-latency (~2 ms) excitatory or inhibitory interactions (Fig. S5). That said, follow up studies with greater Ns are needed to parse out these details

> ***Reviewer #2 (Recommendations for the authors):***
>
> *(1) While I appreciate displaying the success rate for the sake of simplifying behavioral data in Figure 1B, it would be nice to also see these data broken out as correct vs incorrect for stay vs shuttle trials, since it is difficult to determine whether the performance increases are primarily driven by mice improving at stay vs shuttle responses*

We appreciate this feedback. In the revised manuscript, we present data separating stay versus shuttle trials (Fig.1; Fig.S2).

> *(2) In Figure 2 the comparison between shuttle and stay is not particularly convincing, since the comparison is also essentially movement vs no movement and place1-->place2 vs place1-->place1. A more appropriate comparison might be action state neurons vs action content neurons during A-->B, B-->A, or both crossings. If it is true that these populations contain this information, then action state neurons should traverse a large component space in both directions, action content neurons only one direction, and so on.*

We agree that the comparison is not ideal due to differences in locomotion. However, it provides valuable information suggesting that the ACC plays a limited role during stay trials, despite these trials involve mental and cognitive processes comparable to shuttle trials. While we appreciate the reviewer's suggestion, the proposed analysis is not particularly reliable given the relatively small number of simultaneously recorded action-state or action-content neurons.

> *(3) I would say the above point applies to Figure 3 as well. I would also note that this reviewer greatly appreciates the rigor of showing ensemble activity in each subject.*

We appreciate this comment. See our response above.

> *(4) In Figure 5 do these neurons show the same A-->B vs B-->A firing patterns during correct vs incorrect shuttles? The text describing the data in Figure 4 suggests this should be the case but even from a quick glance it sort of seems like the population dynamics during correct vs incorrect shuttles are not the same. My concern is that averaging neural activity over 5s windows washes out all these dynamics*

Preliminary analysis suggests that these firing patterns apply to both correct and incorrect shuttles. However, the main reason we did not compare correct and incorrect trials is the limited amount of data. In many sessions, there are only a few (≤5) incorrect shuttles, which include both A→B or B→A shuttles (Fig.1C; Fig.S2), thus lacking the statistical power for a meaningful comparison.

> *(5) Some information on classifier validation is required - was this leave-out validation and if so how many trials were left-out vs tested? K-fold, and if so, how many folds? Was the trial order shuffled for each simulation? Classifiers will pick up within-session temporal information. In addition to this classifier accuracy during the different time points should be compared by a non-parametric test, and compared to the 95th percentile of the label-shuffled distribution.*

Yes, we use standard 10-fold cross-validation. We appreciate the suggestion on trial-order shuffling, and implementing this procedure does not change our original conclusion. Additionally, we have applied a non-parametric test.

> *(6) How exactly were neurons classified as content vs state? Was it the average activity during the 5s following the shuttle? If this is stated I could not really find it easily so I might suggest clarifying.*

We now use a new method for classification of the two neuron types (Fig.7). We have included detailed methods in the revised manuscript.

> *(7) Movement drives cortical neuron activity more than anything else I have ever seen. Really, more than anything else, it would be nice to demonstrate that it is not movement alone or movement multiplexed with place/sensory information/direction driving these responses.*

We have analyzed ACC neuronal activity in relation to locomotion speed. Our results indicate that only a small fraction of ACC neurons (<15%) show speed-correlated activity (Fig.5). It remains unclear whether these speed-related neurons represent a distinct subpopulation within the ACC or reflect recordings from nearby motor cortex. Postmortem examination of the recording sites suggests that most neurons were recorded from the ACC, while a small subset may be located at the border between the ACC and motor cortex. Therefore, it is possible that the small fraction of speed-related neurons originated from the motor cortex.

Furthermore, we identify two distinct groups of ACC neurons: <iaction-state and action-content neurons, both of which tend to show sustained activity even when the animals remain immobile after completing shuttle behaviors. This prolonged activation in the absence of movement suggests that their activity is not directly driven by locomotion. Moreover, action-content neurons are selectively engaged in only one of the two shuttle categories, either rooms A→B or B→A shuttles. Therefore, differences in neuronal activity are unlikely to reflect locomotor differences, given that both shuttle types involve similar movement patterns.

> *(8) In addition to the above, the place-field analysis in Supplemental Figure 5 only shows 4 neurons. Was the whole population analyzed? Is it possible to decode place from the population during the ITI? The data in this figure sort of look exactly like place fields - many cortical neurons and also some hippocampal neurons have more than 1 place field*

We have now provided additional place-field analysis. A comparison with hippocampal CA1 neurons (recorded during the same task) suggests that ACC neurons encode limited spatial information.

> *(9) "a simple Pavlovian association strategy is unlikely to be sufficient for learning the task" ... is Pavlovian occasion setting not a simple association? Tones and contexts both readily act as Pavlovian occasion setters. Similarly positive/negative patterning might also explain how the task is learned.*

We appreciate this comment and have revised the sentence accordingly. It is possible that animals use multiple strategies to learn and perform the task effectively. In the early stages, animals may rely more heavily on sensory–spatial integration, whereas in later stages, sensory- or location-related Pavlovian associative strategies may contribute to performance, particularly when animals begin to show place preferences during inter-trial intervals.

> *(10) I might suggest softening this language and others like it. For example, 2x2 factorial designs are not really novel.*

We have revised the language used to describe the task.

> *(11) Some of the color-scale bars and figures do not have labels. For example, Supplementary Figure 3, Supplementary Figure 5. Please add labels.*

We have added the missing labels to all color bars.

> ***Reviewer #3 (Recommendations for the authors):***
>
> *(1) Some relevant papers that should be cited:*
>
> *https://doi.org/10.1523/JNEUROSCI.4450-08.2008*
>
> *10.1016/j.neuron.2018.11.016*
>
> *https://doi.org/10.1016/j.jphysparis.2014.12.001*

We appreciate these suggestions.

> *(2) Where can we download the data and code?*

We will upload the essential data and MATLAB code to GitHub to accompany the publication of the final version of this paper.

https://doi.org/10.7554/eLife.105774.2.sa0